SIEMENS

MAGNETOM

Software System

Software Installation syngo MR A35 (NUMARIS/4 VA35A)

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Document Version

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1.1 syngo MR system software

1.1.1 Introduction

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1.1.2 Prerequisites

To configure the syngo-based MR system, training in the following areas is required in addition to this software manual:

- syngo training/syngo MR training
- Network training (TCP/IP)
- DICOM training

All passwords required to install and configure the MR system are available in the "Password List" in "CS Knowledge Base" http://smskb.medical.siemens.com/selfservice, with the exception of the service key (variable password), which is dealt with by the regional service.

1.1.3 Abbreviations and definitions

EDB	Event Database
SRS	Siemens Remote Service
QDB	Quality Assurance Database
COBR	CO Backup/Restore tool
Recovery image	Data set containing the data for generating clones of the backed-up system
CSSI	Customer Specific Software Image
n.a.	Not applicable

2.1 Overview

2.1.1 Software installation procedure for the MRC

For syngo MR A35, the **COBR (CO Backup/Restore)** tool is used to create recovery image DVDs and reinstall software.

2.1.2 Software installation procedure for Imager

For syngo MR A35, the **COBR (CO Backup/Restore)** tool is used to reinstall software.

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2.2 Introduction to COBR

When the MR system is set up in the factory, the COBR tool is used to create the recovery image set for the system (all disks). This image set is supplied with the system. No additional Numaris DVD sets will be supplied.

The first disk in a recovery image DVD set contains the COBR tool.

The COBR tool enables CSEs to **create**, **verify**, and **restore** recovery images of MR systems.

2.2.1 Restore

The software is installed at the factory prior to delivery. Service engineers do not normally need to install software on-site, but in some cases reinstallation will be necessary. In these cases, it is possible to restore all disks, or disk C only, or even disk C and other disks separately from different recovery images using the COBR tool.

Use this method to restore the system to its default settings using recovery image DVDs created at the factory.

If the recovery image DVDs have been created on-site, the system will restore to the state that was present when you created the recovery image DVDs.

- **Silent restore method for disk C**

  Using this method will only restore disk C.

  For detailed steps, see (→ Case 1: Restoring disk C using the disk C recovery image DVDs / Page 11).

- **Silent restore method for the system (all disks)**

  Using this method will restore all the disks (C, D, E).

  For detailed steps, see (→ Case 2: Restoring the system using the system recovery image DVDs (all disks) / Page 15).

- **Restoring CSSI system images**

  In some software upgrade cases, if a CSSI image DVD set is delivered with the system, it may be necessary to restore the whole system using the CSSI.

  For detailed steps, please follow the relevant UI.

2.2.2 Create

You can use recovery image DVDs you have created to restore the system the next time this is necessary.

The image DVDs can be created for the whole system (all disks) or for disk C only.

For detailed steps, see (→ Creating recovery images / Page 22).

- **Silent backup of disk C**

  Following system installation, configuration, and adjustment at the customer's site, service engineers can use this method to back up the installed and configured system. Disk C includes all site-specific data.

- **Manual backup of the system (all disks)**

  The recovery image DVDs for the system (all disks) restore all the disks, including the database information on the other disks.

2.2.3 Verify

Once the images have been created, they should be verified using the COBR tool. Confirm recovery image DVDs can be used.

For detailed steps, see (→ Verifying the recovery images / Page 31).

- **Verifying disk C recovery images**
- **Verifying system recovery images**

2.2.4 Comparison with Dynamic Backup/Restore

Recovery image DVDs created on-site include both software and all site-specific data. However, it is still possible for service engineers to back up/restore the site-specific data separately for troubleshooting or service purposes. (For details of backup/restoration procedures, see the next chapter (→ Site-specific data / Page 44)).

If the software has been updated in a way which cannot be covered by a dynamic backup procedure - for example, if a service pack or application package has been installed - COBR is the preferred tool for creating system images for on-site systems. The COBR tool can be used to restore the software in its entirety, including all modifications.

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2.3 Reinstalling software on the MRC with recovery images

There are four main steps in the reinstallation process:

1. **Check the BIOS setting**
2. **Boot the computer**
3. **Restore the recovery image DVDs**
   - Restore disk C using the disk C recovery image DVDs
   - Restore the system using the recovery image DVDs for all disks
4. **Follow-up work**

2.3.1 Step 1. Check the BIOS setting.

Preparations

BIOS CD

Procedure

The BIOS update optimizes the computer settings for NUMARIS/4 and sets the NUMARIS/4 BIOS password and boot device order.

Load the BIOS defaults for the host before the software installation.

1. Reboot the system (power up the host).
2. During the reboot process, press **F10** (for HP host) or **F2** (for Celsius host) repeatedly until BIOS Setup appears. Enter the BIOS password.
3. Check the date/time and correct if necessary. To avoid problems with licenses, installations in Asia should change the installation date by going back 3 days.
4. Go to the **Boot Tab**.
5. Highlight “Boot Device Priority” and press **Enter**.
6. Move the CD-ROM entry to the top of the list.
7. Insert the BIOS-CD into the DVD/CD-ROM drive.
8. Save and reboot.
9. The BIOS-Update will run automatically.
10. Remove the BIOS-CD once it has been automatically ejected.
11. Restart the system by pressing **Ctrl**-**Alt**-**Del**.

2.3.2 Step 2. Boot the computer

Preparations

COBR tool DVD (the first DVD in a recovery image DVD set)

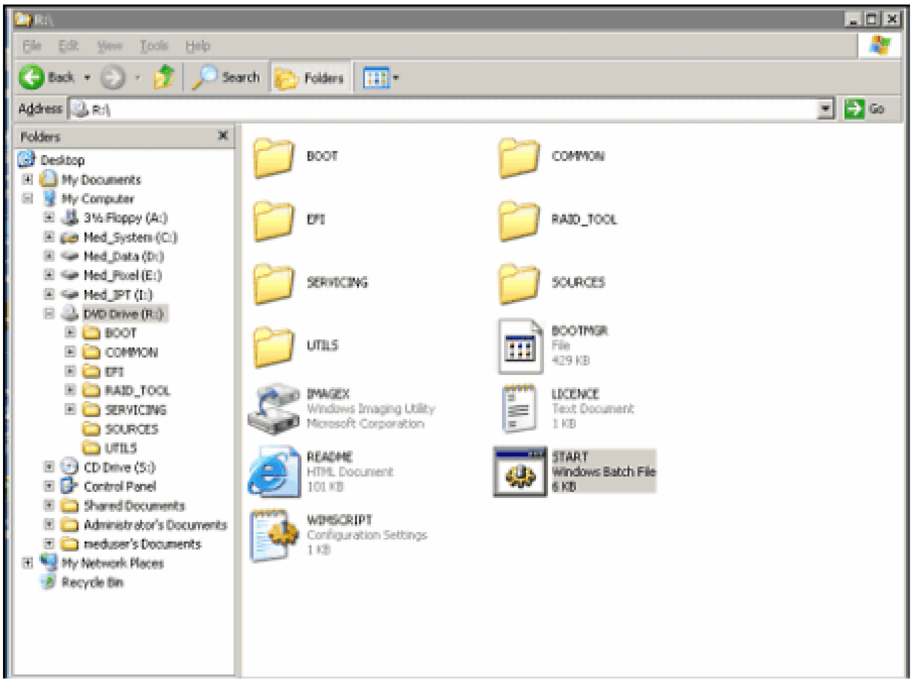
Procedure

There are different booting procedures for bootable and non-bootable computers.

- If the computer is bootable:

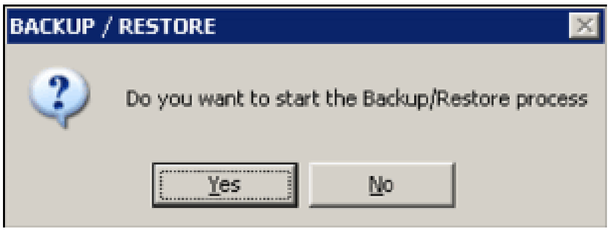
  a) Reboot the computer.
  b) Insert the COBR tool DVD into the CD/DVD-ROM drive R:\.
  c) Log in as `administrator`.
  d) Open Windows Explorer and navigate to drive R:\.
     Double click the “start.bat” program to run it.

Fig. 1: COBR backup/restore start



  e) The message “Do you want to start the Backup/Restore process?” pops up. Click **Yes** to confirm.

Fig. 2: Backup/Restore pop-up



  f) The system restarts. The message “Windows is loading files” is then displayed.

If the computer is not bootable:

  a) Insert the COBR tool DVD into the CD/DVD-ROM drive R:\.
  b) Reboot the computer.

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c) The system restarts. The message “Windows is loading files” is then displayed.

2.3.3 Step 3. Restore the recovery image DVDs

2.3.3.1 Case 1: Restoring disk C using the disk C recovery image DVDs

Preparations

- Recovery Image DVDs of disk C

Function

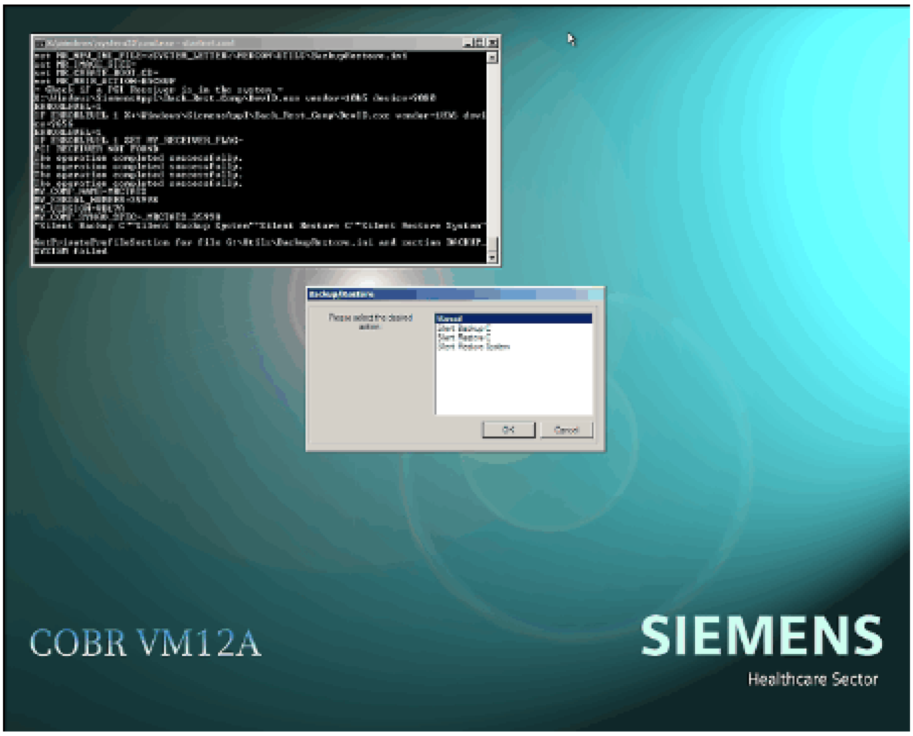
Restoring disk C to the state that was present when you created the recovery image DVDs

Procedure

To restore the latest disk C image to the same computer, use the **Silent Restore C** method.

1. After “Windows is loading files” appears, the COBR user interface is displayed.
2. The Backup/Restore window is displayed.

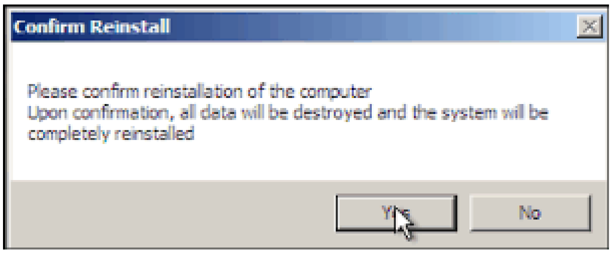
Fig. 3: COBR tool window



3. Select **Silent Restore C**.

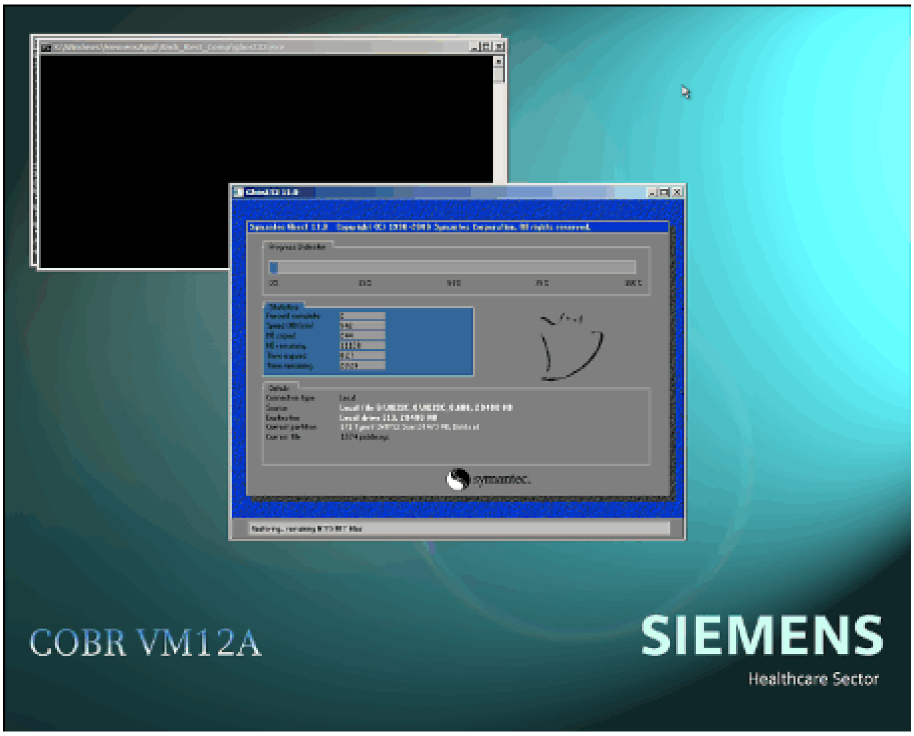
4. If the original software is still valid on the host, a window will pop up at this point to confirm that reinstallation of the software is to take place. Click **OK** to confirm.

Fig. 4: Confirm Reinstallation pop-up



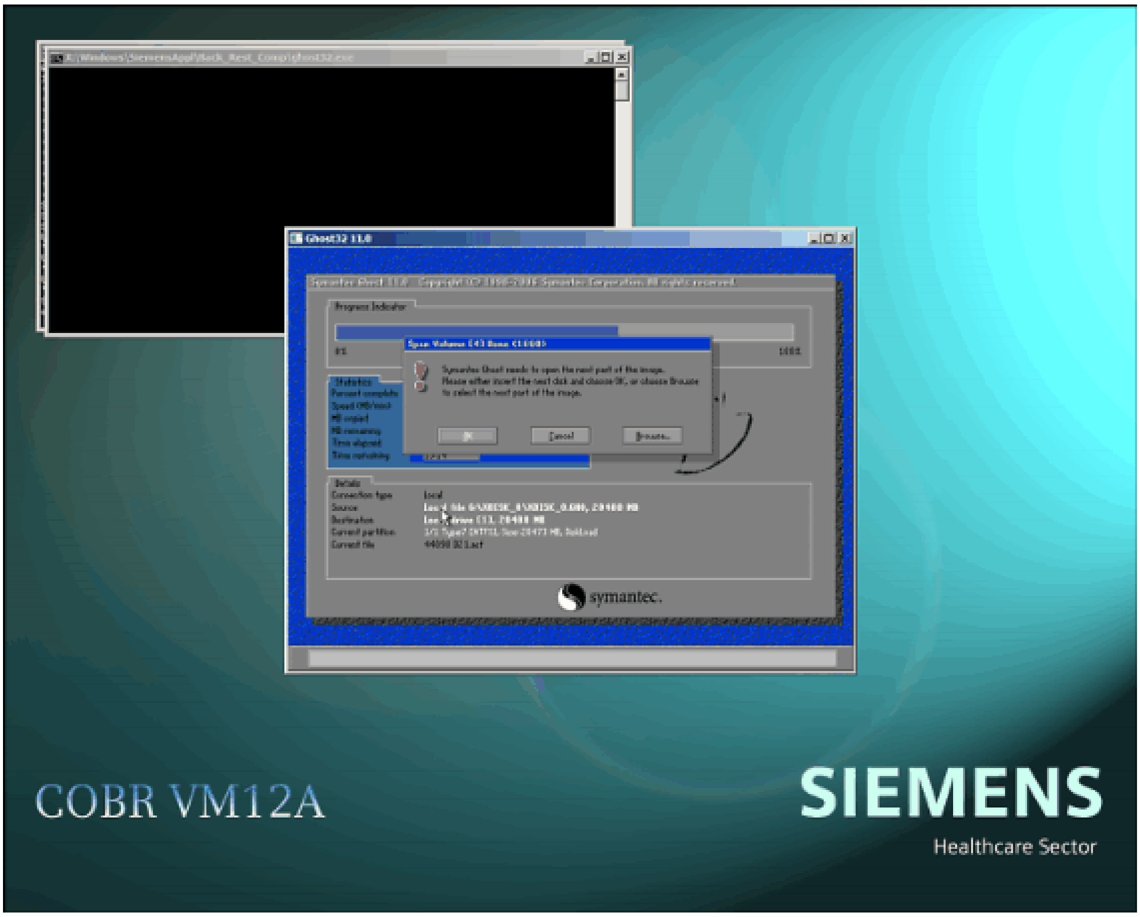
5. Ghost window opens.

Fig. 5: COBR ghost screen restoration



6. "Symantec Ghost needs to open the next part of the image. Please either insert the next disk and choose OK, or choose Browse to select the next part of the image." is displayed.

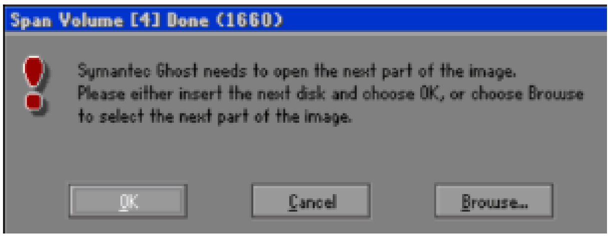
Fig. 6: COBR screen output - Request next DVD



Remove the first DVD from the CD/DVD-ROM drive, insert the next recovery image DVD, and close the drive.

Click **OK** to continue.

Fig. 7: Ghost pop-up request next DVD



This step will be repeated until all data is restored.

7. “Restore of the first disk is finished successfully. Please remove the DVD. To reboot press the 'OK' button” is displayed. (Note: here “the first disk” refers to disk C).

Fig. 8: Backup/Restoration finished successfully

Remove the DVD and close the drive.

Click **OK**.

8. Windows restarts.
9. After rebooting, autologin as `administrator`.
10. Once you are logged in, some configuration tasks will be performed automatically. Wait until these have finished. The system will automatically reboot again.

    During these steps, do not close any command windows or confirm any popup windows:

Fig. 9: Displayed pop-ups 1

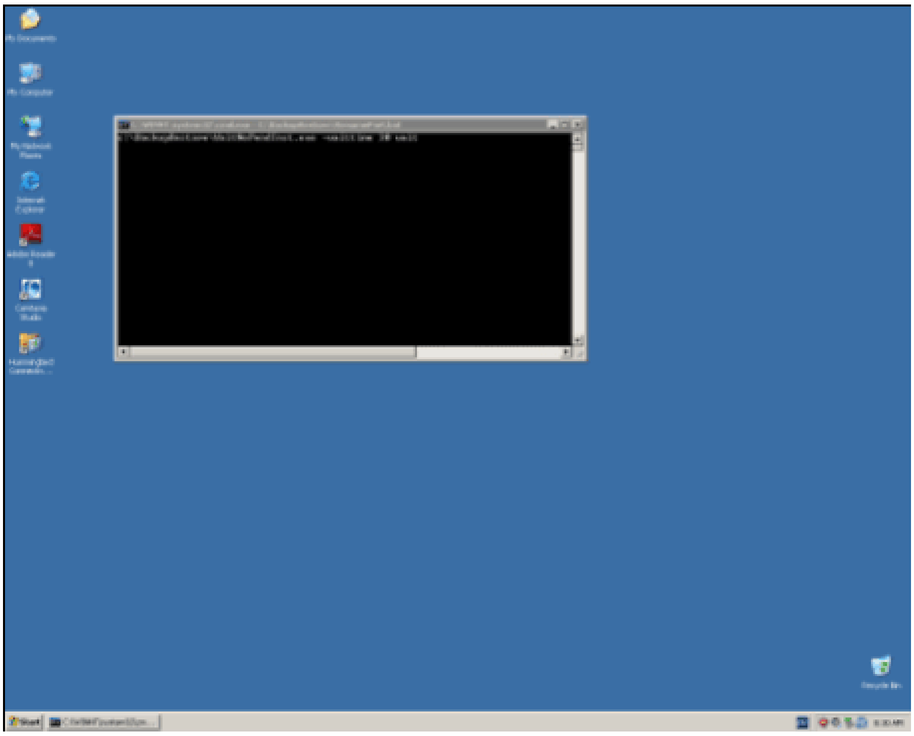
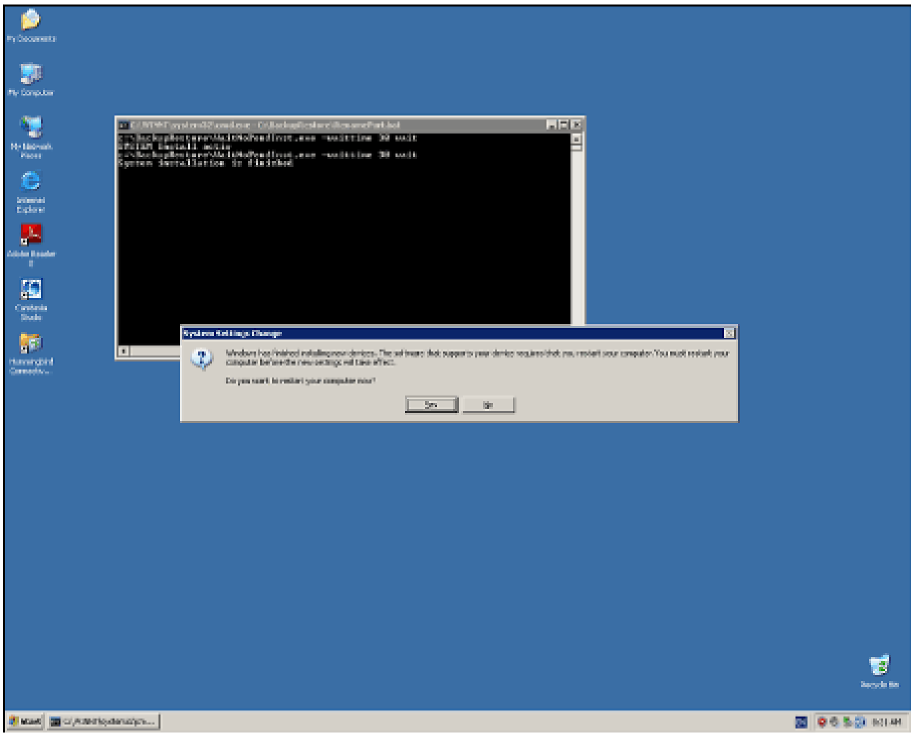


Fig. 10: Displayed pop-ups 2



11. The system is rebooted.
12. If autologin is disabled, log in as `meduser` in the login window.
13. Wait until the main UI is displayed. Restoration has now finished.

The restoration procedure (Silent Restore C) takes around 25 minutes.

2.3.3.2 Case 2: Restoring the system using the system recovery image DVDs (all disks)

Preparations

- Recovery Image DVDs of system (all disks)

If the system recovery image DVD set was created at the factory, the system will be restored to the settings that were made during installation at the factory. If the system recovery image DVD set was created on-site, the system will be restored to the same state as when it was backed up.

Function

Restoring the system to the state that was present when you created the recovery image DVDs (including disks C, D, and E)

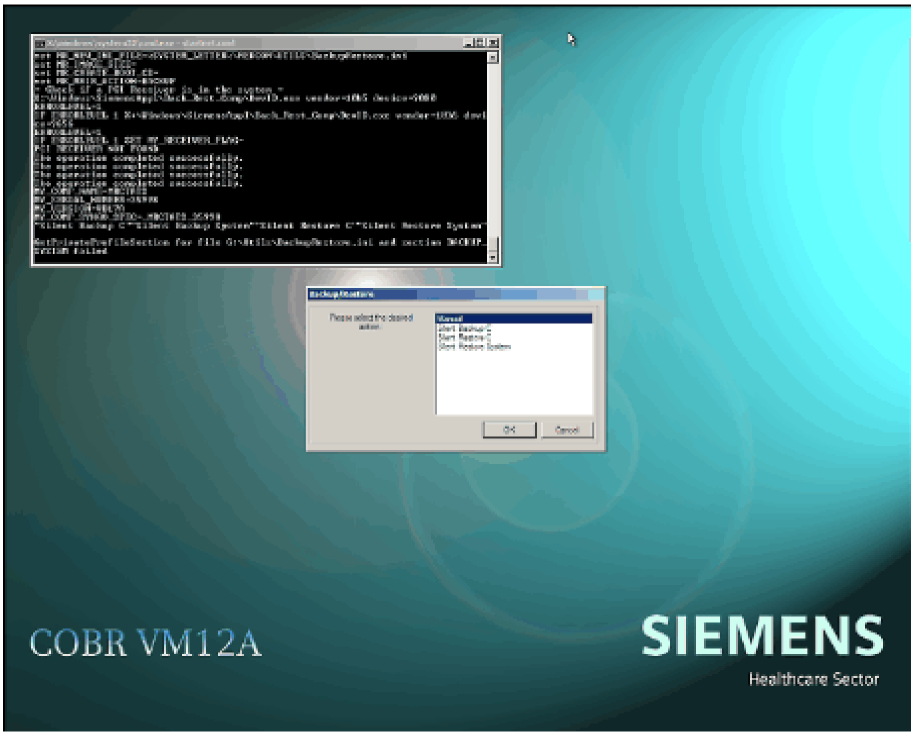
Procedure

To restore system images, use the **Silent Restore System** method.

1. After “Windows is loading files” is displayed, the COBR tool will boot and the COBR user interface will appear.

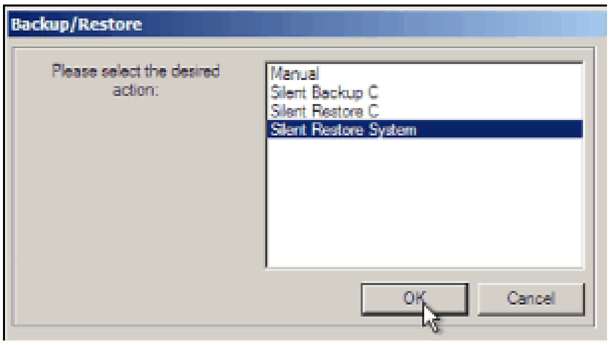
2. The Backup/Restore window is displayed.

Fig. 11: COBR tool window



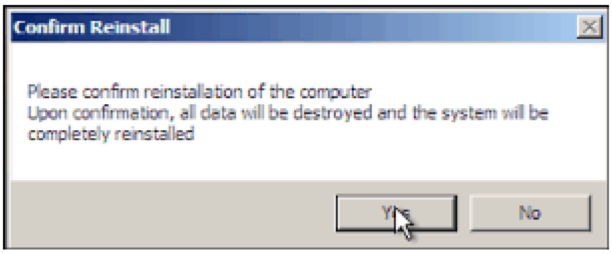
3. In the Backup/Restore window, select **Silent Restore System** and click **OK**.

Fig. 12: Backup/Restore pop-up Silent Restore System



4. If the original software is still valid on the computer, a message window will pop up asking you to confirm that reinstallation is to take place. Click **Yes** to confirm.

Fig. 13: Confirm Reinstallation pop-up



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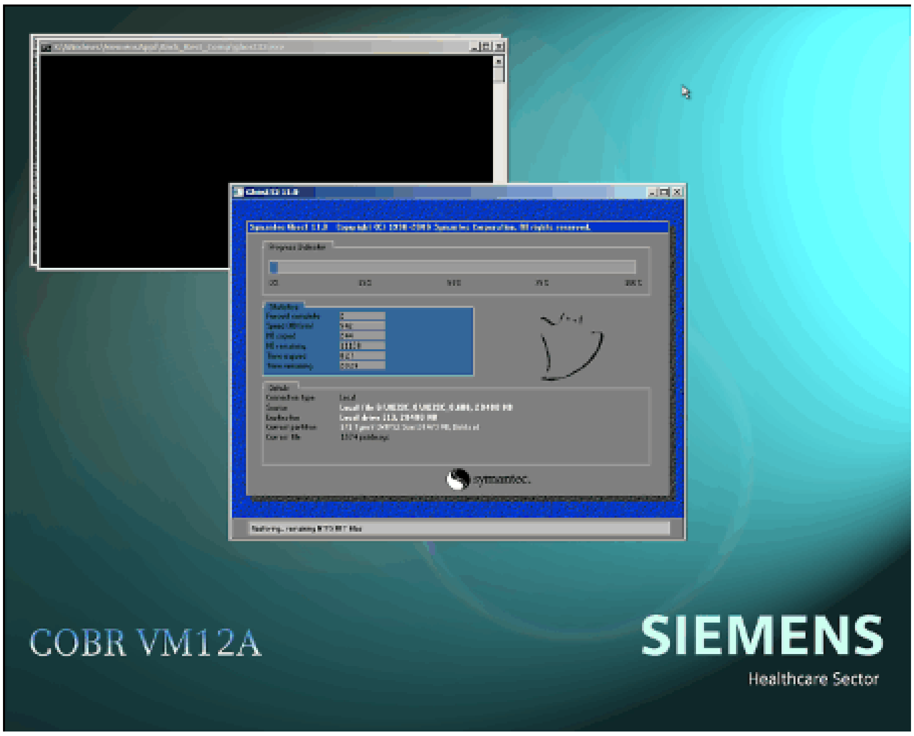
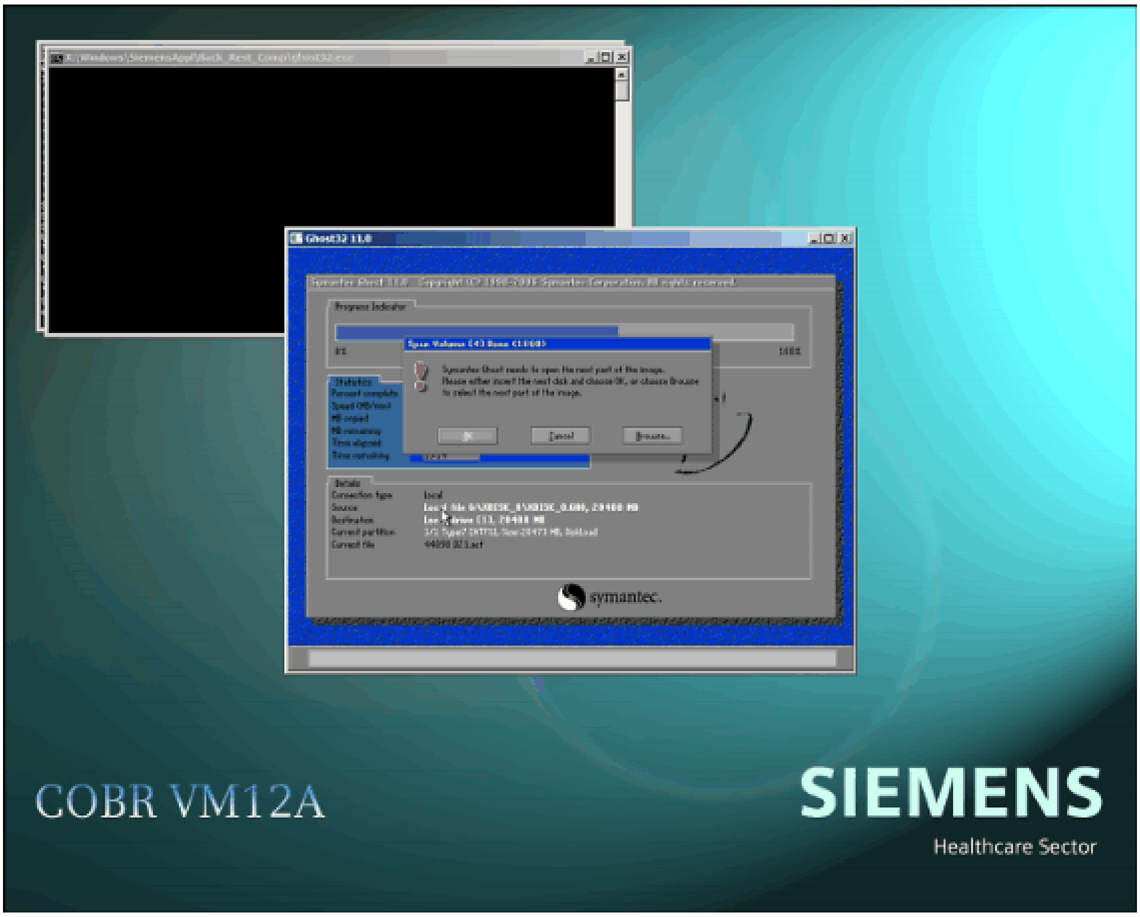
5. Ghost window opens.

Fig. 14: COBR ghost screen restoration



6. If the system image DVD set contains more than one DVD, “Symantec Ghost needs to open the next part of the image. Please either insert the next disk and choose OK, or choose Browse to select the next part of the image.” will be displayed.

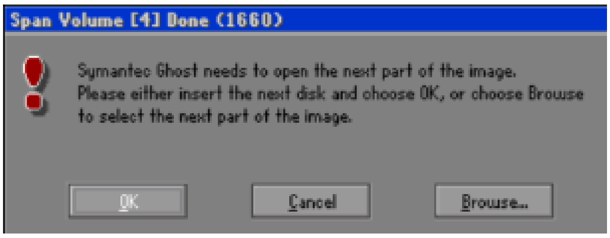
Fig. 15: COBR screen output - Request next DVD



Remove the ejected DVD from the CD/DVD-ROM drive, insert the next DVD from the system image DVD set, and close the drive.

Click **OK** to continue restoration.

Fig. 16: Ghost pop-up request next DVD



This procedure will be repeated until all data is restored.

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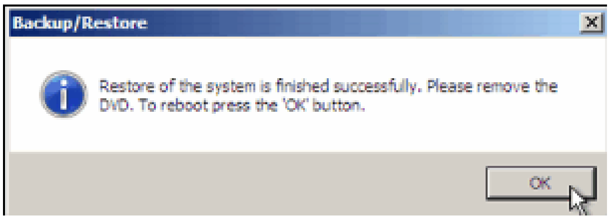
7. Once all data has been successfully restored, “Restore of the system is finished successfully. Please remove the DVD. To reboot press the 'OK' button” is displayed.

Fig. 17: Backup/Restore - Successfully finished pop-up



Remove the ejected DVD and close the drive.

Click **OK** to reboot.

8. The system reboots.
9. After rebooting, autologin as `administrator`.
10. Following autologin, some configuration tasks will be performed automatically. Wait until these have finished. The system will automatically reboot again.

During these steps, do not close any command windows or confirm any popup windows:

Fig. 18: Displayed pop-ups 1

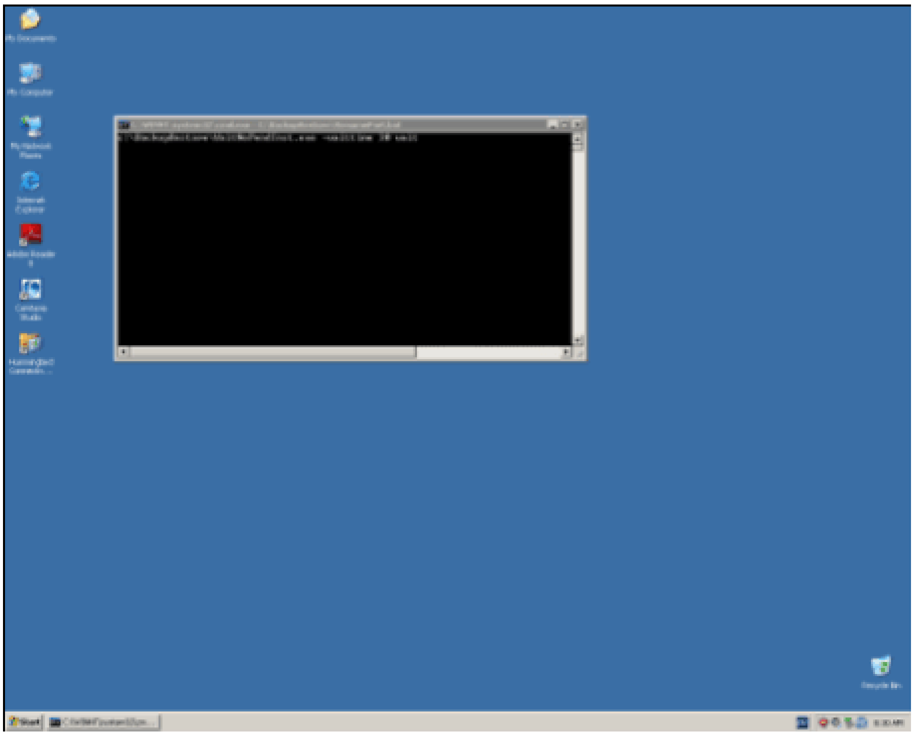
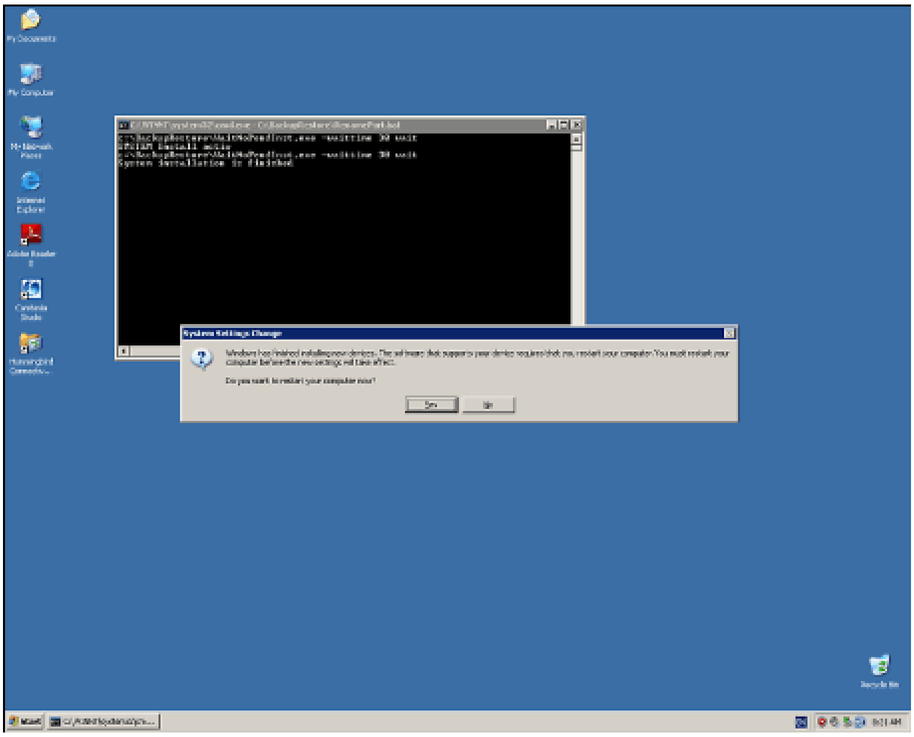


Fig. 19: Displayed pop-ups 2



11. After rebooting has taken place, log in as `meduser` if autologin is disabled.
12. Wait until the main UI is displayed. Restoration is now complete.

The restoration procedure takes around 30 minutes.

2.3.4 Step 4. Follow-up work

- If the system has been restored with recovery images that were created at the factory and do not contain any site-specific data, then the follow-up work outlined below will need to be carried out:

1. Reboot MARS. Wait until the system is ready.
2. Check whether the relevant update instructions require any service packs or hotfixes for this software version. If they do, follow the update instructions to install them.
3. Restore the site data; see (→ Site-specific data / Page 44).
4. Once you have restored the site-specific data and/or installed new service packs, you can start up and operate the system. It is recommended that you create new recovery image DVDs; you can then use these to restore the system the next time this is necessary. For information on how to create new recovery image DVDs, please refer to (→ Creating recovery images / Page 22).

NOTICE

Once the software has been installed, check the information in the main menu under Help - Info to verify the software baseline label.

The correct software label for the syngo MR A35 release is:

» COEM VE10E

» syngo VE31G

» syngo VX69A

» N4_VA35A_LATEST_20091210

» N4_VA35A_LATEST_20091210_P1

2.4 Creating recovery images

At the customer site, backing up disk C is recommended as an efficient way of backing up the installed system with all its configuration settings, as disk C contains all software, site-specific data, and so on. Use the **Silent Backup C** method to do this.

Preparations

1. COBR tool DVD

   The COBR tool can be found on the first disk of a recovery image DVD set. The recovery image DVD set from the factory or the most recent one created on-site may be used.

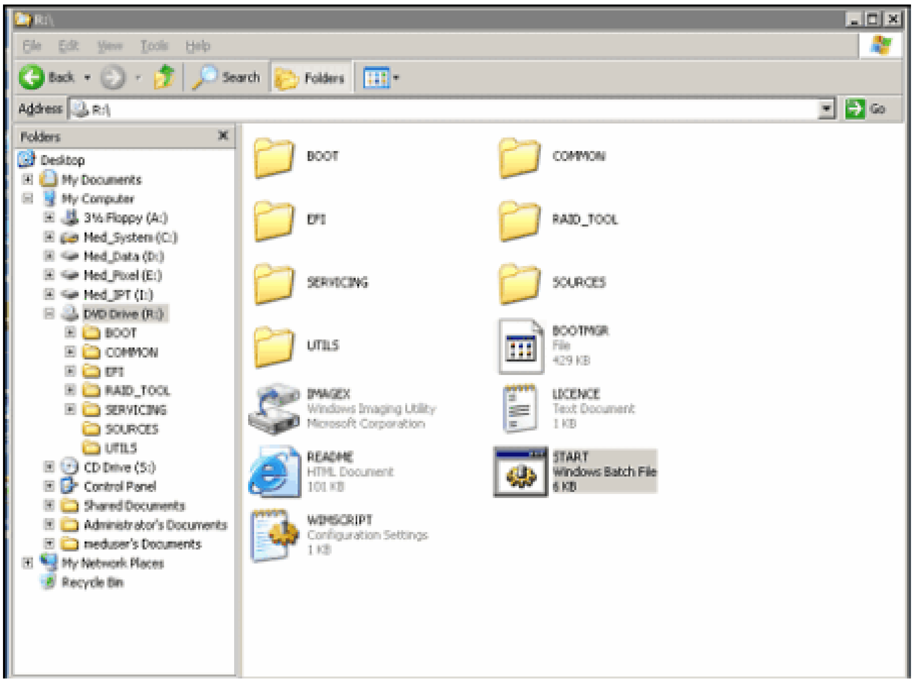
2. Blank DVDs

Procedure

1. Turn on the computer.
2. Insert the COBR tool DVD into the CD/DVD-ROM drive.
3. Log in as `administrator`.
4. Open Windows Explorer and navigate to the CD/DVD-ROM drive **R:**.

   Double click the “start.bat” program to run it.

Fig. 20: COBR backup/restore start



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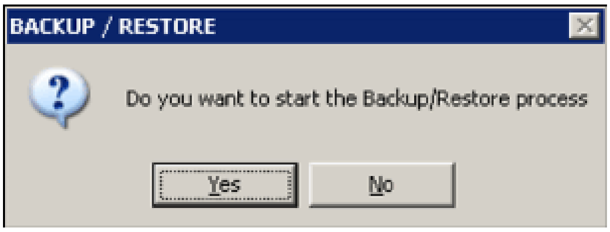
5. A message pops up: "Do you want to start the Backup/Restore process?"

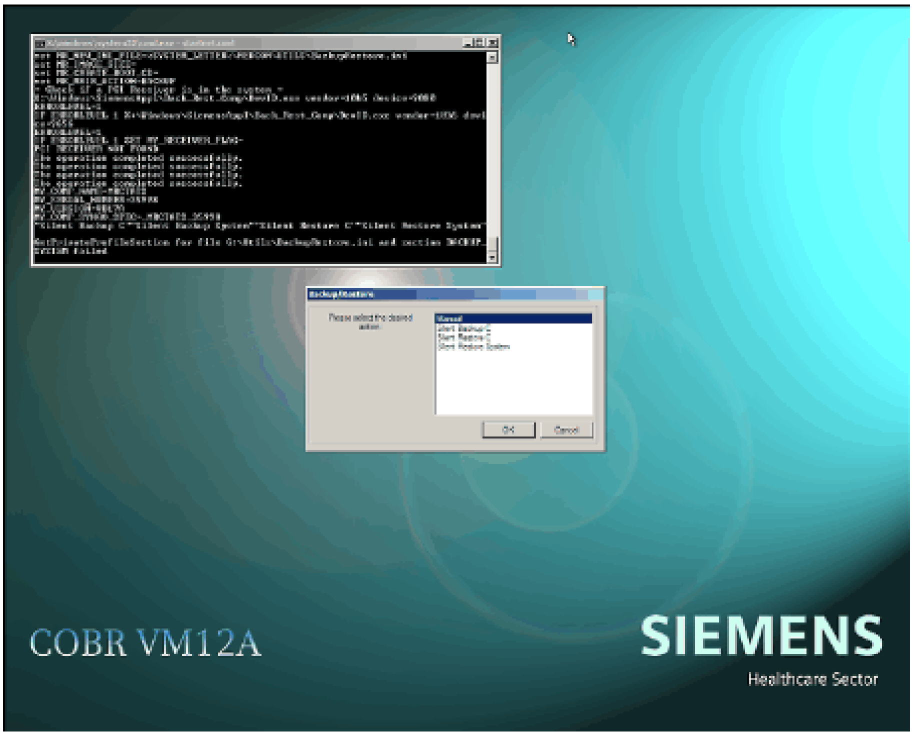
Fig. 21: Backup/Restore pop-up



Click **Yes** to confirm.

6. The computer restarts and the COBR tool boots.
7. Wait until the COBR user interface is displayed.

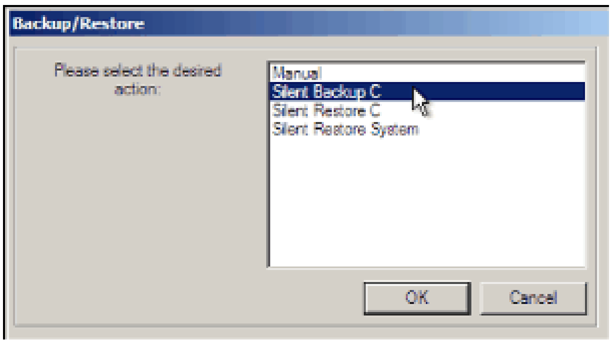
Fig. 22: COBR tool window



8. Creating recovery images for **disk C**

   a) In the Backup/Restore window, select **Silent Backup C** and click **OK**.

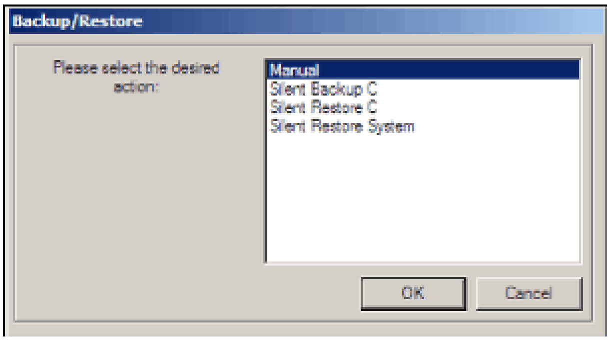
Fig. 23: Backup/Restore - Silent Backup C



Creating recovery images for **the system (all disks)**

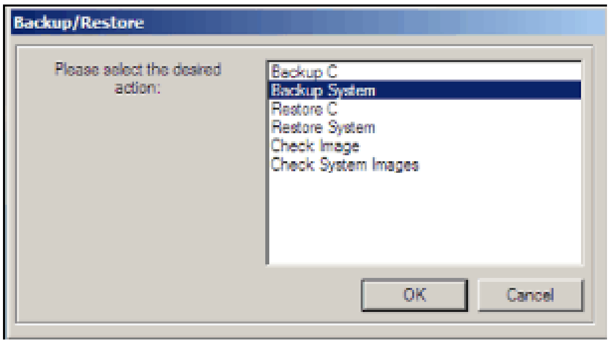
a) In the Backup/Restore window, select **Manual** and click **OK**.

Fig. 24: Backup/Restore - Manual



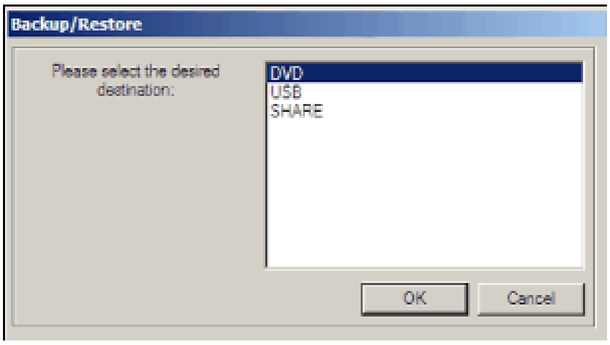
b) In the next Backup/Restore window, select **Backup System** and click **OK**.

Fig. 25: Backup/Restore - Backup System



c) In the next Backup/Restore window, select **DVD** as the medium for the backup image and click **OK**.

Fig. 26: Backup/Restore - DVD



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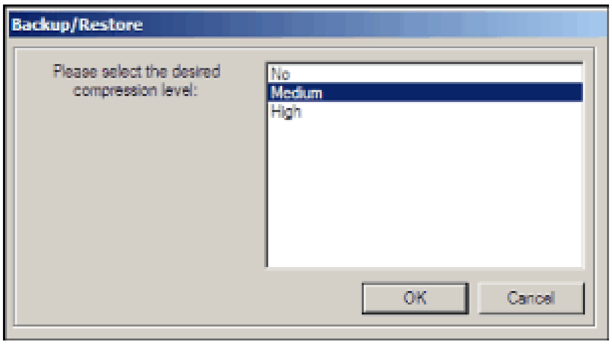
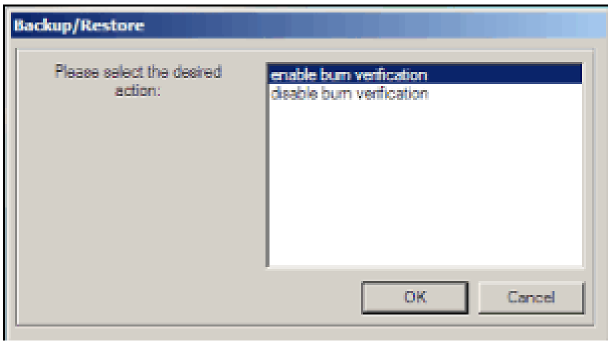
d) In the next Backup/Restore window, select **Medium** as the compression level and click **OK**.

Fig. 27: Backup/Restore - Compression level



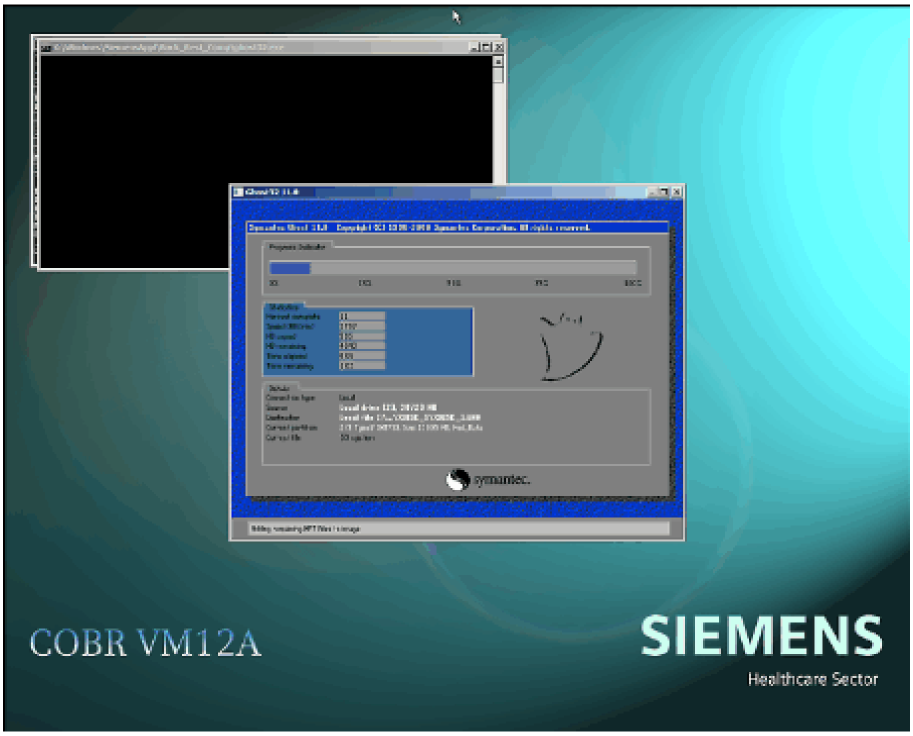
e) In the next Backup/Restore window, select **enable burn verification** and click OK.

Fig. 28: Backup/Restore - Enable burn verification



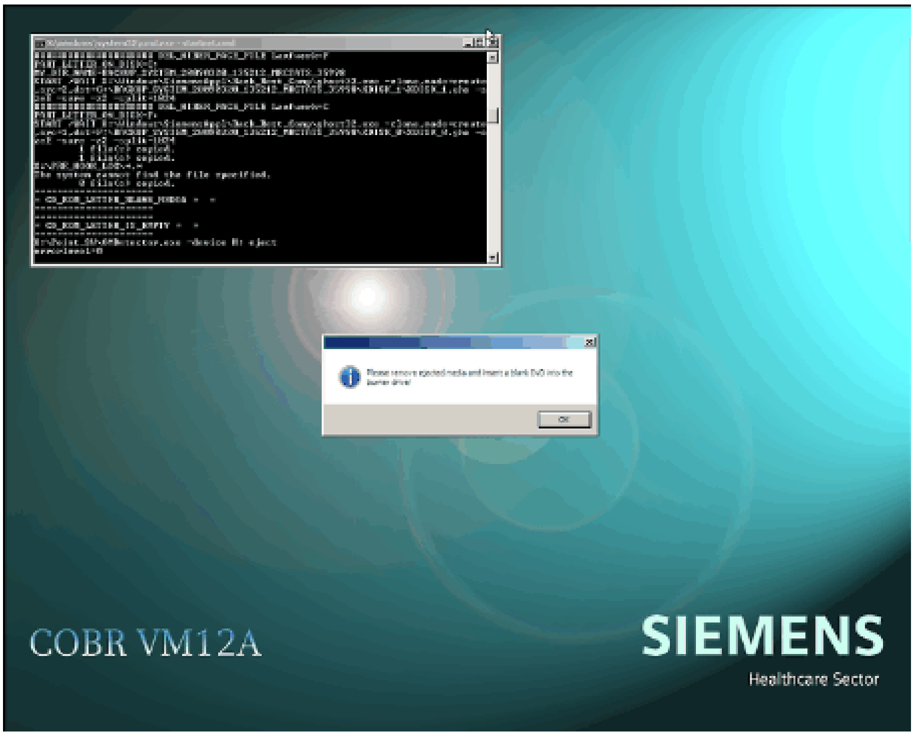
9. A ghost window appears and the backup process starts.

Fig. 29: COBR ghost screen



10. The message “Please remove ejected media and insert a blank DVD into the burner drive!” pops up.

Fig. 30: COBR screen output - Request DVD

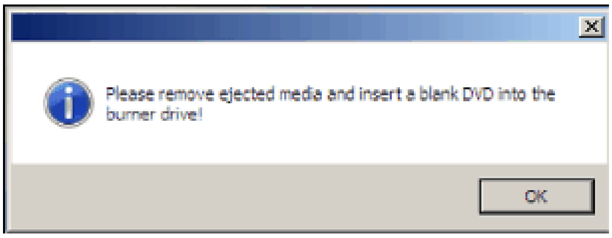


Remove the ejected DVD from the CD/DVD-ROM drive.

Insert a blank DVD into the CD/DVD-R drive and close the drive.

Click the **OK** button on the message window.

Fig. 31: Blank DVD pop-up



11. The system starts to create the image DVD.

    The screenshots below show some examples.

Fig. 32: DVD generation ISO file

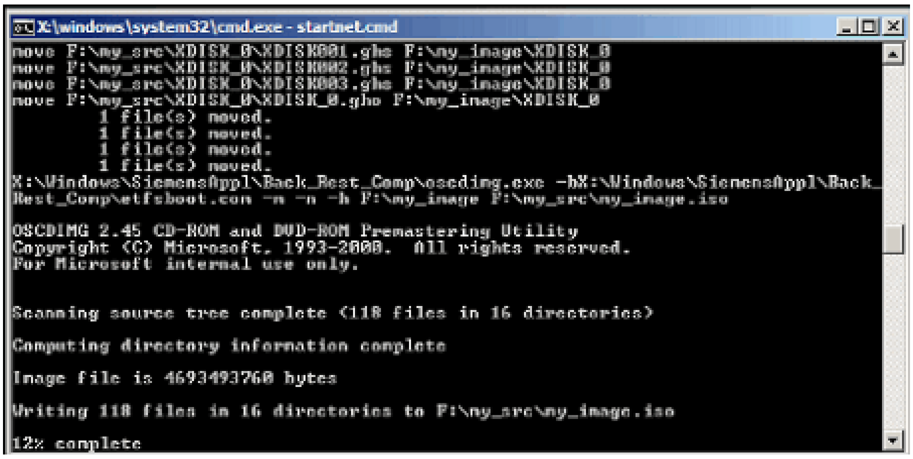


Fig. 33: DVD generation - Writing data to DVD

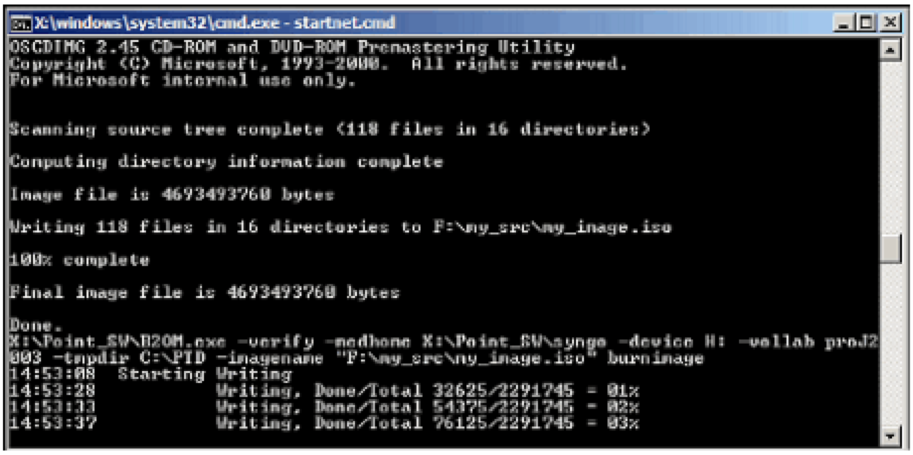
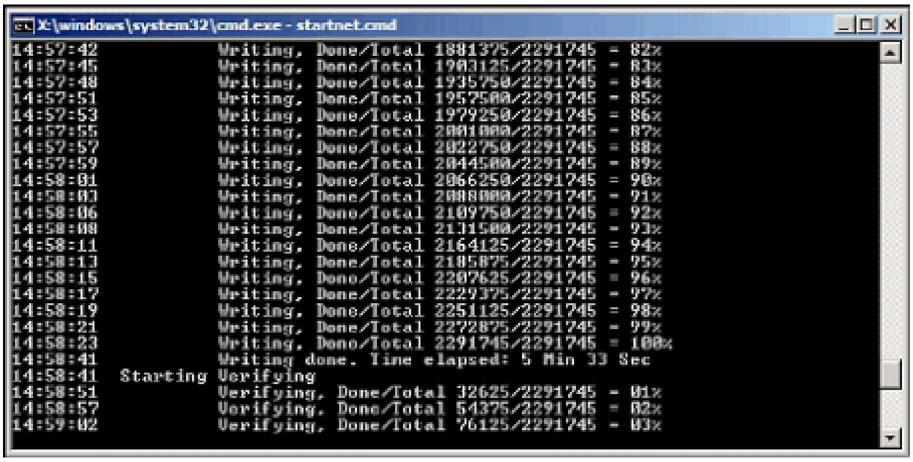
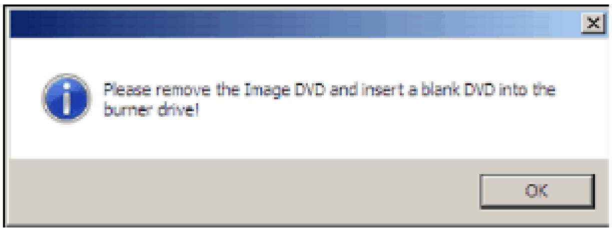


Fig. 34: DVD generation - Verifying DVD



12. If another blank DVD is required, the message “Please remove the Image DVD and insert a blank DVD into the burner drive!” will pop up.

Fig. 35: DVD generation - Request pop-up



Remove the ejected DVD and label it.

Insert a blank DVD and close the drive.

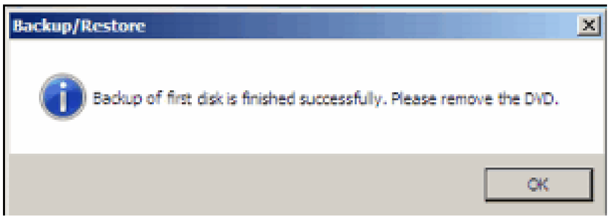
Click the **OK** button on the message window.

This image DVD is created in the same way as the first one.

This procedure will be repeated until all data is written to the DVDs.

13. Once all data has been successfully written to the DVD, the message “Backup of first disk is finished successfully. Please remove the DVD.” will be displayed. (Note: here “Backup of first disk” refers to the backup of disk C).

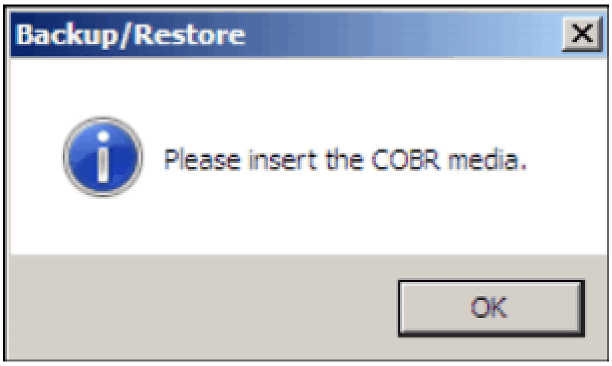
Fig. 36: Backup/Restoration finished successfully



Remove the ejected DVD, close the drive, and click **OK**.

14. The message “Please insert the COBR media” pops up.

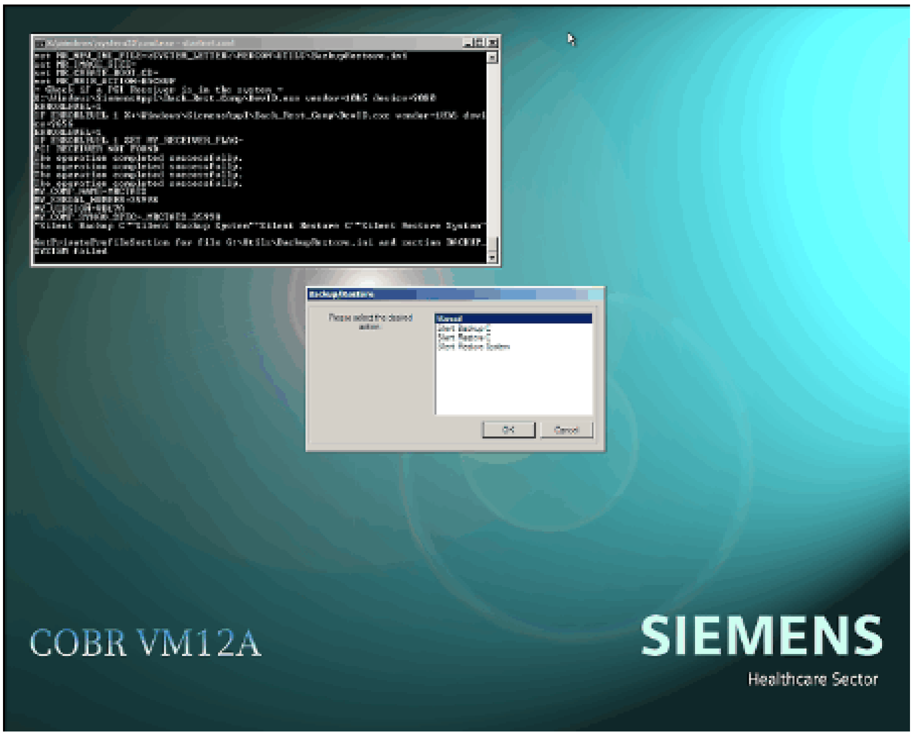
Fig. 37: Backup/Restore - Request COBR tool



Insert the COBR tool DVD into the CD/DVD-ROM drive, close the drive, and click **OK**.

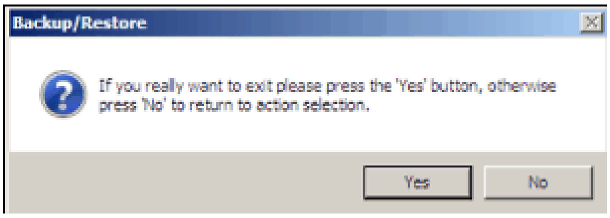
15. The main Backup/Restore window appears. In this window, click **Cancel** to exit.

Fig. 38: COBR tool window



16. A message pops up to confirm that you wish to exit.

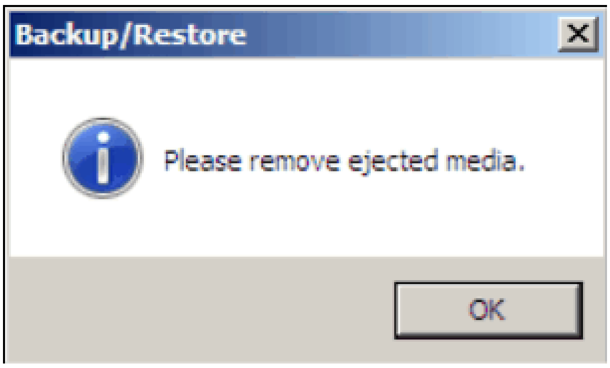
Fig. 39: Confirmation pop-up



Click **Yes** to confirm.

17. The message "Please remove ejected media." pops up.

Fig. 40: Backup/Restore - Remove ejected medium pop-up



Remove the ejected DVD from the drive and close the drive.

Click **OK**.

18. The system restarts automatically.

The process of creating the recovery image DVD set for disk C (Silent Backup C) takes about 45 minutes.

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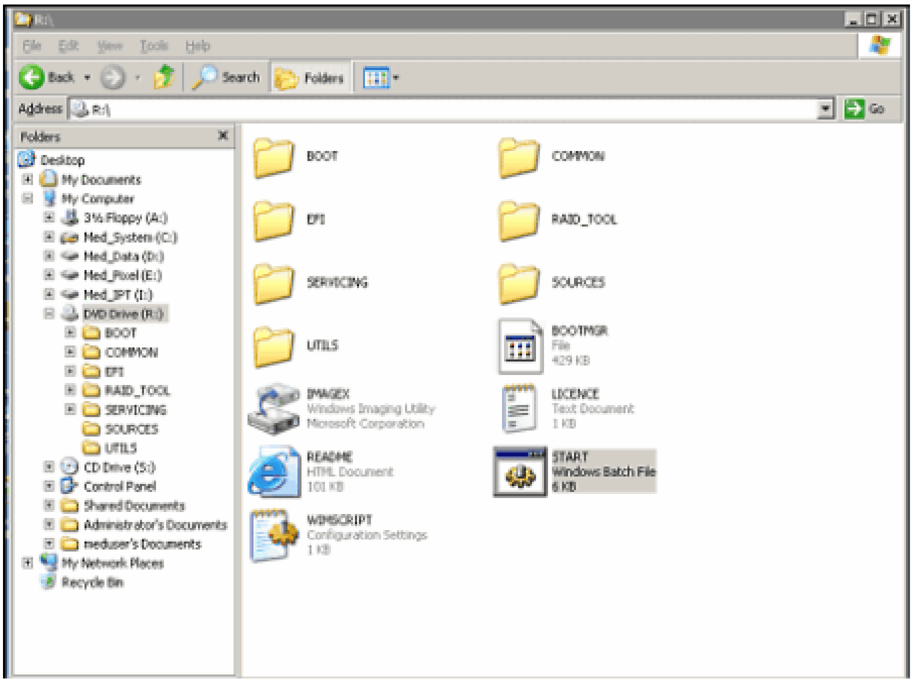
2.5 Verifying the recovery images

After the recovery image DVDs for disk C have been created, verify the DVDs using the following procedure:

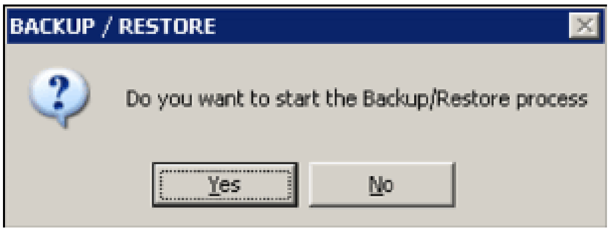
1. Reboot and log in as `administrator`.
2. Insert the first DVD from the most recently created disk C image DVD set into the CD/DVD-ROM drive.
3. Start Windows Explorer and navigate to the CD/DVD-ROM drive R:\.
4. Find the file “start.bat” and double-click to run it.

Fig. 41: COBR backup/restore start



5. A message pops up: “Do you want to start the Backup/Restore process?”

Fig. 42: Backup/Restore pop-up

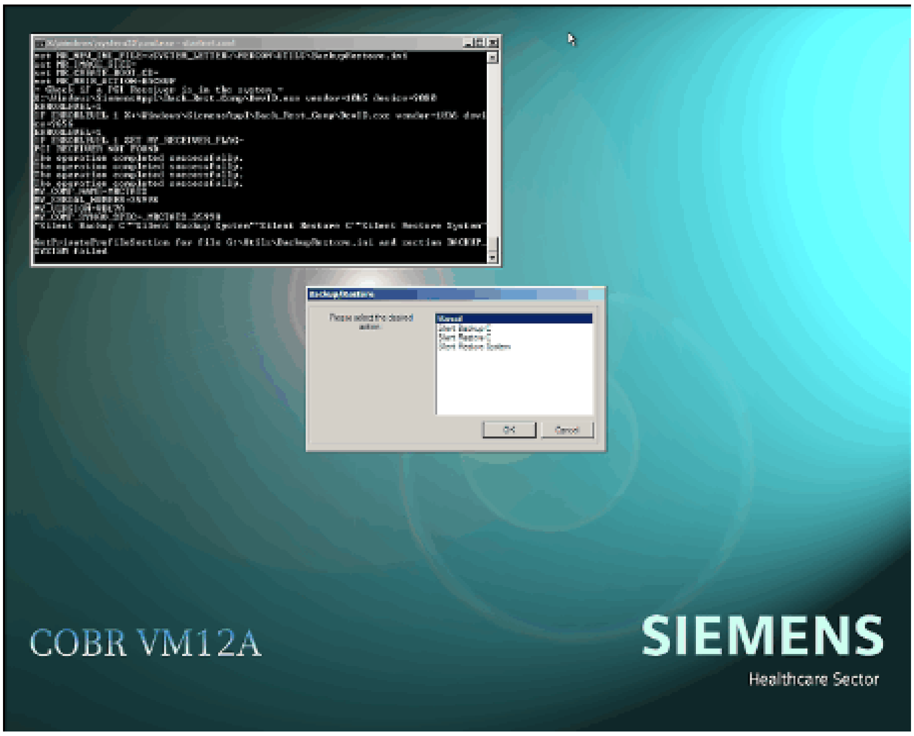


Click **Yes** to confirm.

6. A reboot is performed and the COBR tool boots.

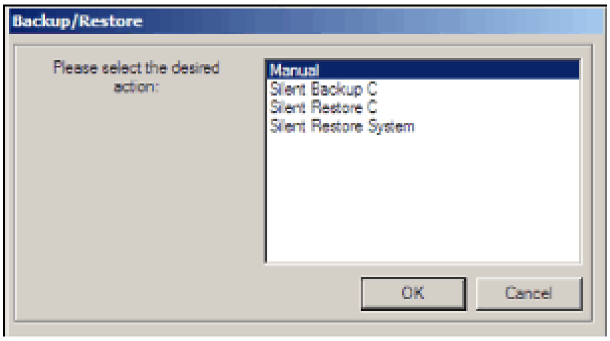
7. Wait until the COBR tool window is displayed.

Fig. 43: COBR tool window



8. In the Backup/Restore window, select **Manual** and click **OK**.

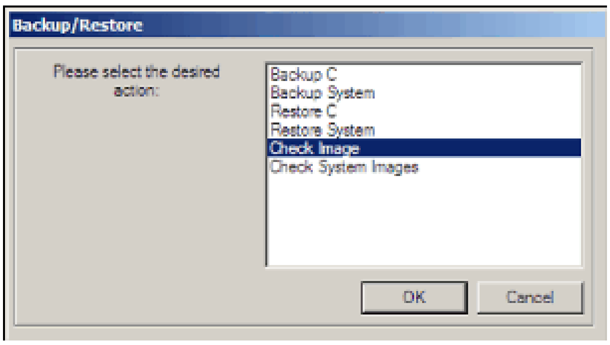
Fig. 44: Backup/Restore - Manual



9. Verifying the recovery image for **disk C**

   a) In the next Backup/Restore window, select **Check Image** and click **OK**.

   Fig. 45: Backup/Restore - Check Image



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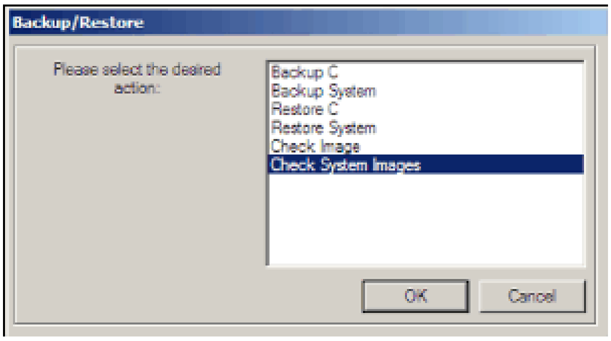
Verifying the recovery image for t**he system (all disks)**

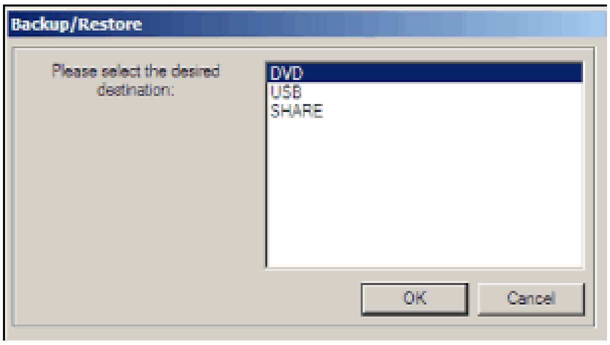
a) In the next Backup/Restore window, select **Check System Images** and click **OK**.

Fig. 46: Backup/Restore - Check System Images



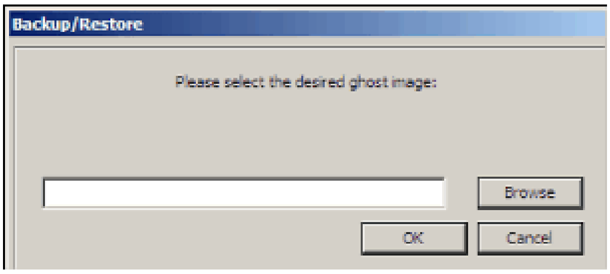
10. In the next Backup/Restore window, select **DVD** and click **OK**.

Fig. 47: Backup/Restore - DVD



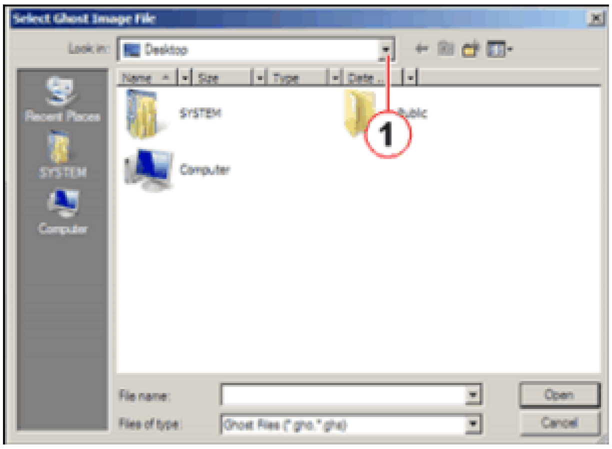
11. In the next Backup/Restore window, click **Browse** to find the target image.

Fig. 48: Backup/Restore select ghost image



12. In the “Select Ghost Image File” window, click the drop-down menu under “Look in:” .

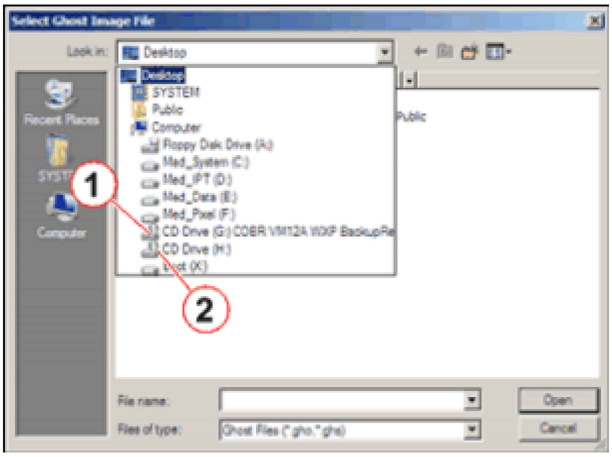
Fig. 49: Select Ghost Image File



(1) Drop-down menu

13. From the drop-down menu, select the drive where the image DVD is inserted; for example, G: or H:.

Fig. 50: Select the drive

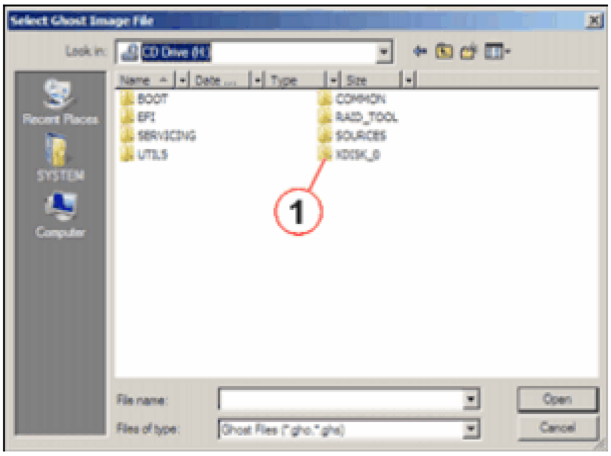


(1) Drive G: (CD/DVD-R)

(2) Drive H: (CD/DVD-ROM)

14. In the image DVD, find the directory `XDISK_0`.

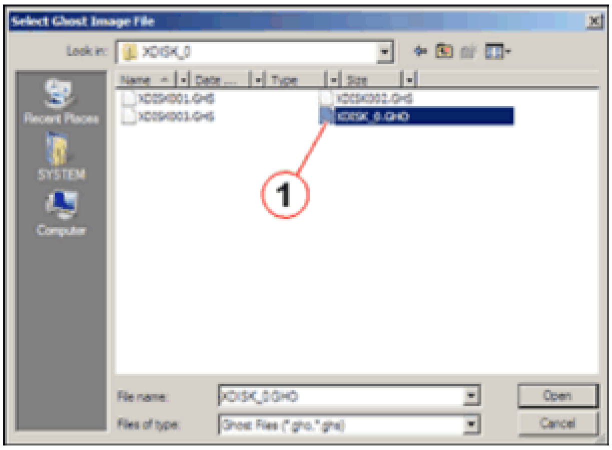
Fig. 51: Select Ghost Image Directory



(1) Directory XDISK_0

15. In the directory `XDISK_0`, find the ghost image file `XDISK_0.GHO`, select it, and click **Open**.

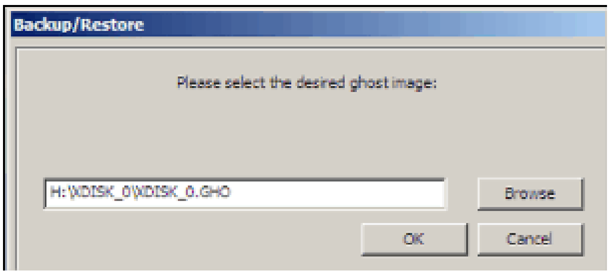
Fig. 52: Select the Ghost Image file



(1) Ghost Image File XDISK_0.GHO

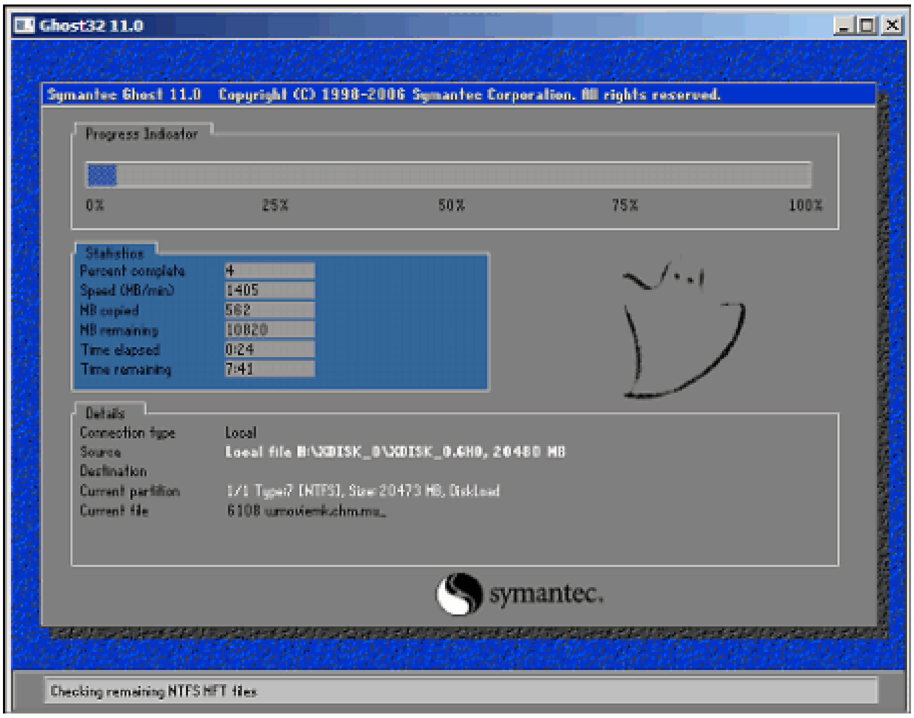
16. In the Backup/Restore window, click **OK** to start the image verification process.

Fig. 53: Backup/Restore pop-up



17. The ghost image verification process will now start.

Fig. 54: Ghost pop-up



18. If the system image DVD set contains more than one DVD, a message will pop up asking for the next DVD: "Symantec Ghost needs to open the next part of the image. Please either insert the next disk and choose OK, or choose Browse to select the next part of the image."

Fig. 55: Ghost request pop-up

Eject and remove the DVD from the drive.

Insert the next DVD and close the drive.

Wait about 10 to 15 seconds for the DVD to be recognized by the drive.

Click **OK** to continue with the verification process.

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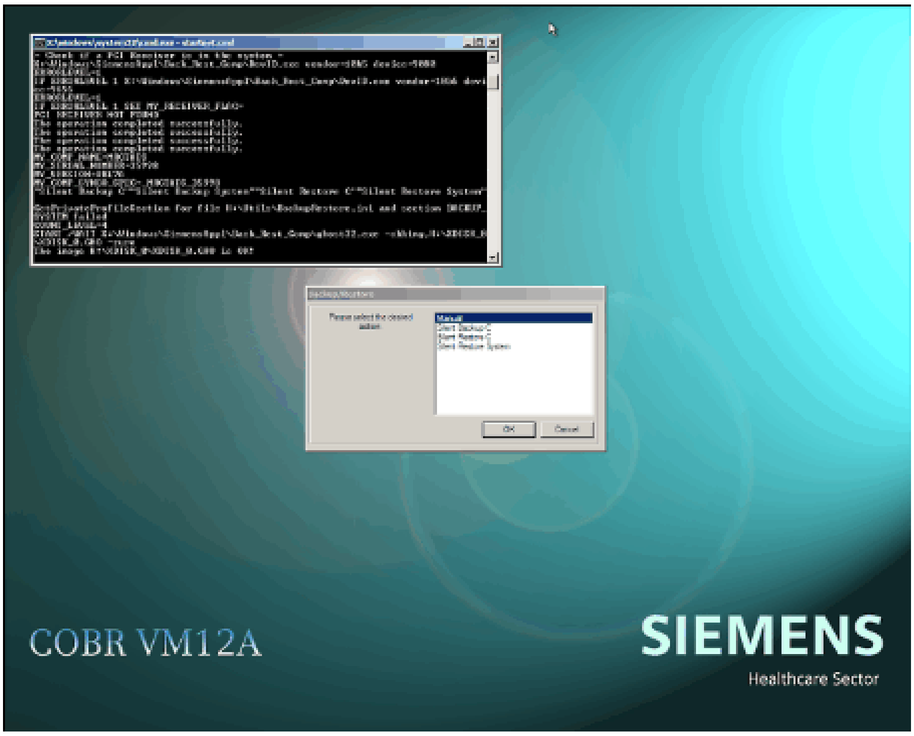
19. Wait until the main Backup/Restore window reappears.

Fig. 56: Ghost image verification - Complete

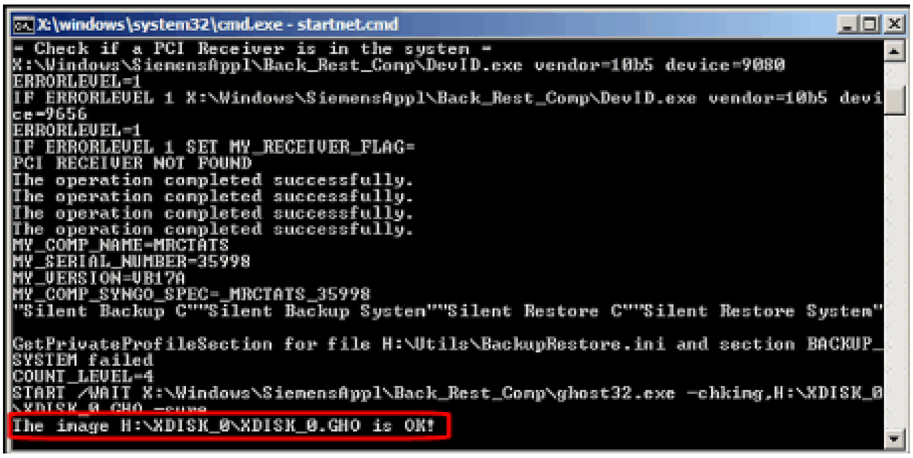


A message should appear in the command tool window:

“The image G:\XDISK_0\XDISK_0.GHO is OK!”

or “The image H:\XDISK_0\XDISK_0.GHO is OK!”.

Fig. 57: Command tool window verify image



Verification is then complete; the image is OK.

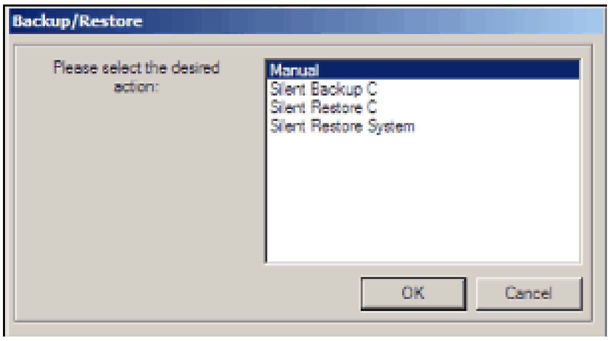
Note:

If the image is not OK, then

a) Perform the image verification again.
b) If the verification process fails again, create the disk C image again (see (→ Creating recovery images / Page 22)).

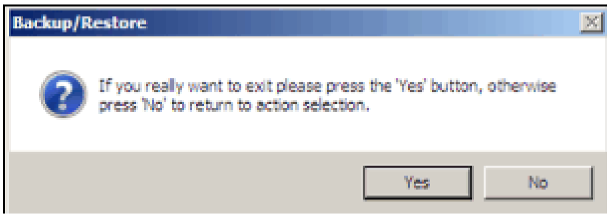
20. In the main Backup/Restore window, click **Cancel** to exit.

Fig. 58: Backup/Restore - Manual



21. A message pops up to confirm that you wish to exit.

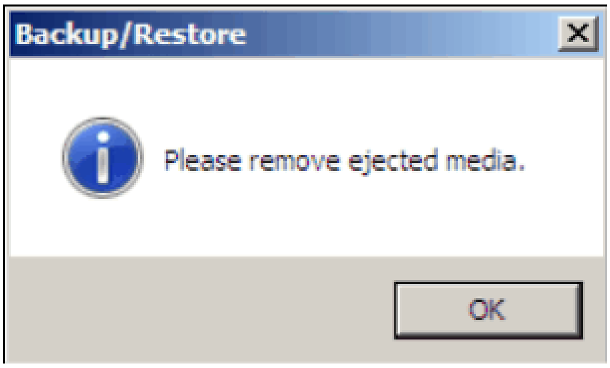
Fig. 59: Confirmation pop-up



Click **Yes** to confirm.

22. The message “Please remove ejected media.” pops up.

Fig. 60: Backup/Restore - Remove ejected medium pop-up



Remove the ejected DVD from the drive and close the drive.

Click **OK**.

23. The system is rebooted.

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This chapter describes how to install the Imager software.

> The MAGNETOM syngo-MR Imager is delivered with factory-installed software. The installation procedure outlined below is only required if the host is being replaced.

3.1 Prerequisites

- MRIR COBR image DVD for MR A35
- The measurement network is connected.

3.1.1 Checking/Importing the Imager license to the MRC

- Log in as administrator (MRC).
- Use Windows Explorer to copy and paste the license file to **C:\Medcom\tftp** on the MRC.
- Rename the license file for Imager to **License.dat**.

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3.2 Installation procedure

> Whenever the error message "LMGRD has encountered a problem and needs to close" appears while you are establishing a remote desktop connection to Imager, please ignore it. It will not affect the system.

1. On the host, go to Start - Run and type the command "showmemrir" in order to establish a remote desktop connection to Imager.

   » If you are then able to connect to Imager, proceed as follows.

   a) Enter user name "MRSystem" and password "MRSystempwd", and log into Imager.
   b) Insert the MRIR COBR image DVD in the DVD-ROM drive.
   c) Open drive R: and double click the start.bat file.
   d) The message "Do you want to start the Backup/Restore process?" pop up. Click **Yes** to confirm.
   e) A reboot is performed and the restoration process starts automatically.
   f) Once the restoration process has successfully completed, the MRIR COBR image DVD will be ejected.
   g) Remove the MRIR COBR image DVD.
   h) Switch the MRIR off.
   i) Switch the MRIR back on.
   j) The MRIR boots up. The missing partition is created and rebooting is performed automatically.
   k) Following rebooting, the missing raw data files are created. This will take about 5 minutes.
   l) The MRIR is now ready for measurement.

   » If you are unable to connect to Imager, proceed as follows.

   a) Insert the MRIR COBR image DVD in the DVD-ROM drive.
   b) Reboot the MRIR. The restoration process will start.
   c) Once the restoration process has successfully completed, the MRIR COBR image DVD will be ejected.
   d) Remove the MRIR COBR image DVD.
   e) Switch the MRIR off.
   f) Switch the MRIR back on.
   g) The MRIR boots up. The missing partition is created and rebooting is performed automatically.
   h) Following rebooting, the missing raw data files are created. This will take about 5 minutes.
   i) The MRIR is now ready for measurement.

3.2.1 Checking connectivity from the host to Imager and Imager raw drives

- On the host, go to Start - Run and type the command "showmemrir" in order to establish a remote desktop connection to Imager. The user name is "MRSystem" and the password is "MRSystempwd".
- Open a cmd tool and check that the "MattauchBuffer" and "rawexchange" devices are available:

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3.3 Final checks

Check whether hotfixes and service packs are available for this software version according to the various speedinfos and update instructions.

- Perform the "Quality Assurance" procedure: "Image Orientation" on the Service Platform.
- Make sure that the system is generating images.

4.1 Backing up site-specific data

Required time: 10-60 min.

Required tools/parts: CD-R

> Only use a CD-R with a medical grade, e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

> The user settings will be restored during the first full login as meduser. After this, you will need to reboot again to activate the restored language settings.

4.1.1 General

You should save site-specific data each time changes are made to the site-specific configuration settings, tune-up specifications, or sequences and protocols. This ensures a valid set of site-specific data can be available for restoration purposes if a hard disk crashes.

There are two methods for backing up site-specific data:

- Master backup
- Conventional backup

Before starting a data backup on the system, restart the MR application using the System Manager, especially if user settings have changed.

The backup must always be executed as meduser.

During a data backup, Windows Explorer and other active editors must be closed, otherwise errors may occur.

If the backup CD is left in the CD writer, an automatic restoration of all dynamic data is performed immediately after software installation.

4.1.2 Master backup procedures

1. Log in as meduser.
2. Start the Service Platform and select **Backup & Restore**.
3. Insert a recordable CD into the CD/DVD-burner.
4. Select **Backup** from the “Command” drop-down menu.
5. In the Drives menu, select the target drive (drive letter **S:**). The data will be transferred to the selected CD/DVD burner.
6. Select "**Master Backup**" in the "**Packages**" menu.

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7. Click **Go** to start the backup process and check for errors in the log window once it is complete.

> If there are errors regarding **"ITO"** during the backup process, check if CA Unicenter has been installed previously. If CA Unicenter has been installed, this is an error. Write down the CA Unicenter configuration settings. If CA Unicenter has not been installed, please ignore this error.

> When the backup process is finished, the CD will be ejected automatically. If an "rti.exe runComp an_Server" error appears on the MRC after a few seconds, it can be ignored.

8. Click the **Home** button to return to the startup page of the Service Platform.

4.1.3 Conventional backup procedures

1. Log in as meduser.
2. Start the Service Platform and select **Backup & Restore**.
3. Insert a recordable CD into the CD/DVD-burner.
4. Select **Backup** from the "Command" drop-down menu.
5. In the Drives menu, select the target drive (drive letter **S:**). The data will be transferred to the selected CD/DVD burner.
6. Depending on the data to be backed up,
   - Select "**SW-Settings02**" in the "**Packages**" menu.
     Click **Go** to start the backup process and check for errors in the log window once it is complete.

> If there are errors regarding **"ITO"** during the backup process, check if CA Unicenter has been installed previously. If CA Unicenter has been installed, this is an error. Write down the CA Unicenter configuration settings. If CA Unicenter has not been installed, please ignore this error.

> When the backup process is finished, the CD will be ejected automatically. If an "rti.exe runComp an_Server" error appears on the MRC after a few seconds, it can be ignored.

   - Select "**Numaris4-Package**" in the "**Packages**" menu.
     Click **Go** to start the backup process and check for errors in the log window once it is complete.
   - Select "**Security Settings-Package**" in the "**Packages**" menu.
     Click **Go** to start the backup process and check for errors in the log window once it is complete.
   - Select the other items to be backed up in the "**Packages**" menu.
     Click **Go** to start the backup process and check for errors in the log window once it is complete.

7. Click the **Home** button to return to the startup page of the Service Platform.

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4.2 Restoring site-specific data

4.2.1 General

The site-specific data can be restored manually following software installation or while the system is running, using the steps outlined below.

4.2.2 Manually restoring master backup data

Having a set of site-specific data on hand following a software installation will save you a lot of work and time. The site-specific data (e.g. configuration, tune-up, and QA settings) is included in the master backup.

1. Log in as **meduser**.
2. Start Internet Explorer.
3. Click the “Click here to start the Local Service Configuration” link in Internet Explorer.
4. Log in to the Service Platform and select **Backup & Restore**.
5. Insert the backup disk in the CD/DVD-ROM (drive letter **R:**).
6. Select **Restore** in the “Command” drop-down menu.
7. Select the source drive CD/DVD-ROM (drive letter **R:**) in the "Drives" menu.
8. Select the xxx_**Master-Backup**_xxxx_xxxx_ok.ar to be restored in the “Archive” menu.
9. In the menu “Groups”, select the necessary packages to be restored.
10. Click **Go** to start the restoration process and check for errors in the log window once it is complete.
11. Click **Home**.
12. Click **OK** to reboot the system.

4.2.3 Manually restoring conventional backup data

Having a set of site-specific data on hand following a software installation will save you a lot of work and time. The site-specific data (e.g. configuration, tune-up, and QA settings) is included in the master backup.

1. Log in as **meduser**.
2. Start “Internet Explorer”.
3. Click the “Click here to start the Local Service Configuration” link in Internet Explorer.
4. Log in to the Service Platform and select **Backup & Restore**.
5. Insert the backup disk in the CD/DVD-ROM.
6. Select **Restore** in the “Command” drop-down menu.
7. Select the source drive CD/DVD-ROM (drive letter **R:**) in the "Drives" menu.

8. Depending on the data to be restored,
   - Select "SW-Settings02" in the "Archive" menu.
     In the "Groups" menu, select the groups to be restored.
     Click **Go** to start the restoration process and check for errors in the log window once it is complete.
   - Select "Numaris4-Package" in the "Archive" menu.
     Select both groups in the "Groups" menu.
     Click **Go** to start the restoration process and check for errors in the log window once it is complete.
     Note: If there is an error message showing "siemensIRU-PCNT-init cannot be restored", this can be ignored.
   - Select "Security-Settings" in the "Archive" menu.
     Select "All" in the "Groups" menu.
     Click **Go** to start the restoration process and check for errors in the log window once it is complete.
   - Select other items from the "Archive" menu as required and restore these using the same process.
9. Click **Home**.
10. Click **OK** to reboot the system.

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5.1 General

For detailed information regarding the configuration of syngo MR systems, refer to **Help** in the Service Software Platform.

If you have changed the configuration within the Service Software Platform and the message "reboot system pending" is displayed in the "status bar", reboot the system as follows: Click **Home** and confirm the reboot message window with **OK**.

Language settings and the configuration UI for option Chorus are described in the Appendix chapter.

Information on how to configure Siemens Remote Service is contained elsewhere in the "Installation and Startup for SRS" document.

The use of non-released combinations of hard-copy cameras and BU products is prohibited and does not meet the defined requirements. For detailed information regarding the CAMERA configuration, refer to the "Camera Info ..." technical documentation applicable to your NUMARIS/4 software version.

The "CCC" Connectivity Competence Center provides a powerful configuration management tool. Go to:

http://cs.med.siemens.de/cms/en/home/

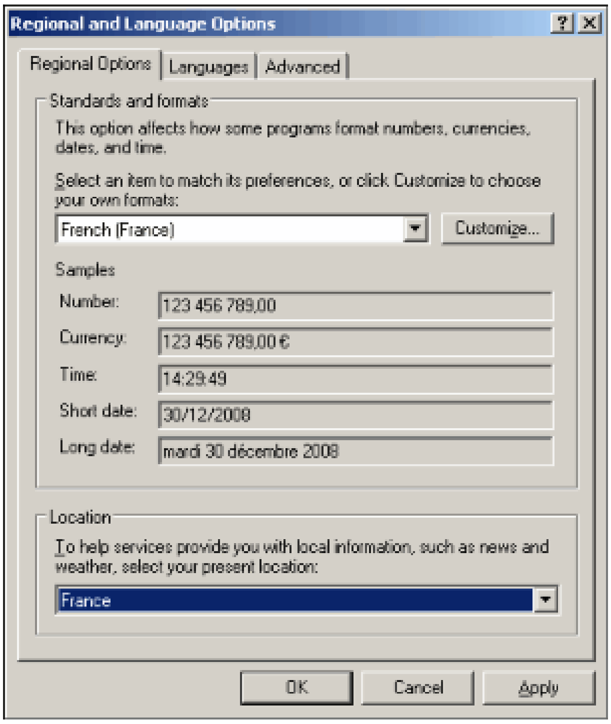
For DICOM Service, on the DICOM Homepage (CCC) navigate to

"Interoperability DB", **Download**, **Acquisition Tool**.

Remember to create a backup if the configuration has changed.

6.1 Language settings

6.1.1 Keyboard and language settings

<table>
<tr><th>ACTION</th><th>MRC</th></tr>
<tr><td>▪ In the main UI, select Options -> Configuration.</td><td>☐</td></tr>
<tr><td>▪ Double click the Regional and Language Settings icon, and open the window:
Fig. 61:

</td><td>☐</td></tr>
<tr><td>▪ On the Regional Options tab, select the appropriate language for the main UI under Standards and formats, using the pull-down menu. Only the following languages can be selected:
- Chinese (PRC)
- English (US)
- English (UK)
- French (France)
- German (Germany)
- Spanish (Spain)
Under Location, choose your country.</td><td>☐</td></tr>
<tr><td>▪ Click Apply.</td><td>☐</td></tr>
</table>

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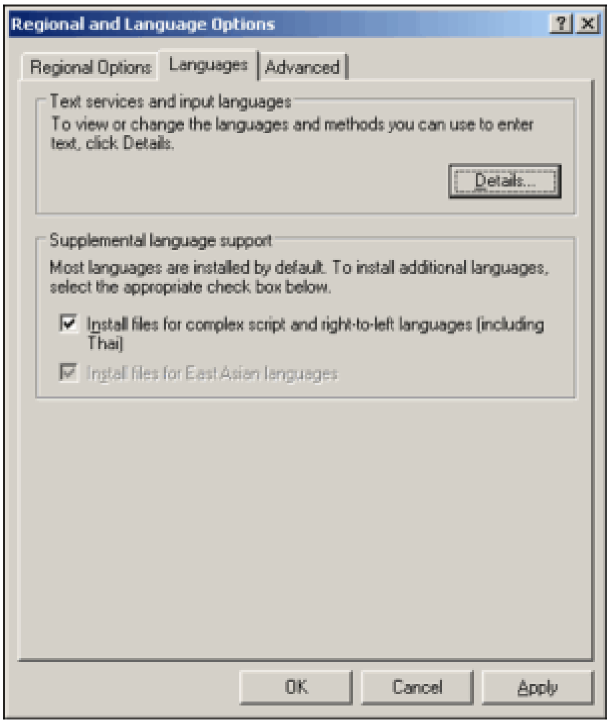
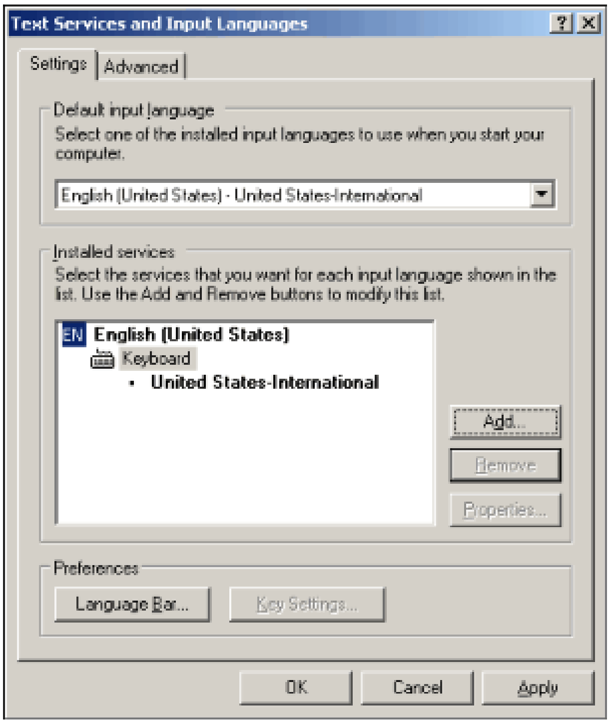
ACTION	MRC
■ In the **Regional and Language Options** window, select the **Languages** tab.	☐

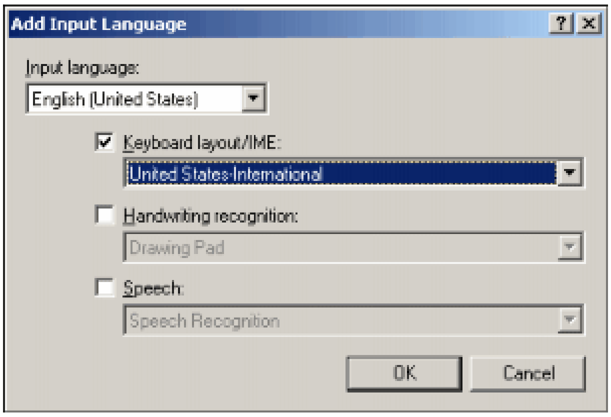
Fig. 62:



ACTION	MRC
■ Click **the Details button**. The **Text Service and Input Languages** window opens:	☐

Fig. 63:



ACTION	MRC
▪ In the **Text Services and Input Languages** window, click the **Add** button. The **Add Input Language** window opens: *Fig. 64:* 	☐
▪ In the window, select the language used for the main UI under **Input language**. Choose the keyboard layout to be used under **Keyboard layout/IME**. Note: The input language determines the measurement system. - English (US) uses pounds as the unit of weight and inches as the unit of length. - English (UK) uses kg as the unit of weight and cm as the unit of length.	☐
▪ Click **OK**.	☐
▪ Click **Apply**.	☐

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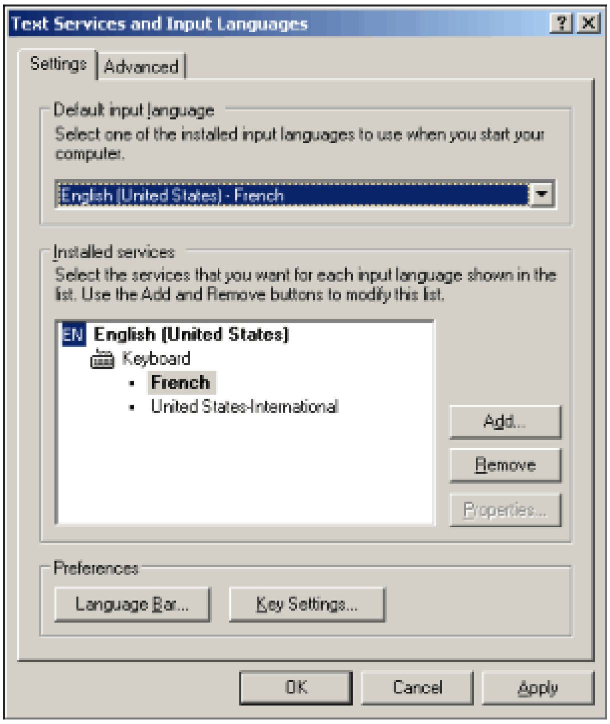
ACTION	MRC
■ In the **Text Services and Input Languages** window, select the newly configured keyboard layout under **Default input language**:	☐

Fig. 65:



ACTION	MRC
■ Click **Apply**.	☐

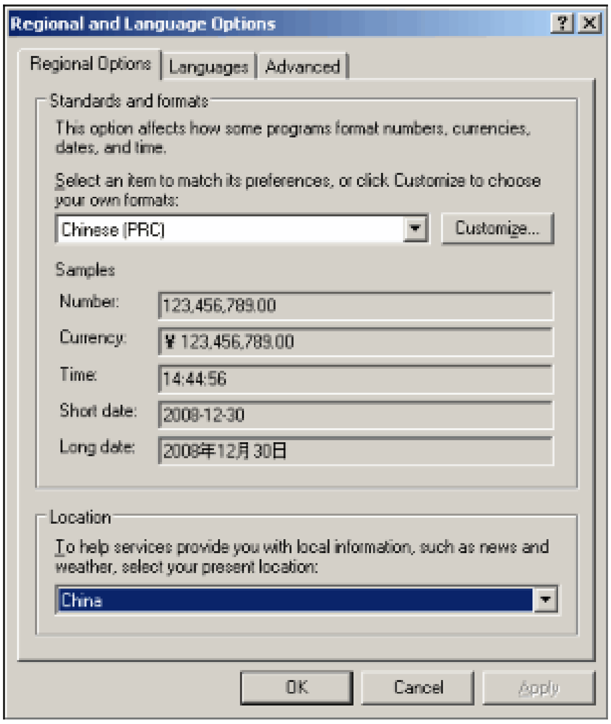
6.1.2 Switching language settings between English and Chinese

6.1.2.1 Switching the language from English to Chinese

ACTION	MRC
■ Log in as meduser. In the main UI, select **Options** -> **Configuration**.	☐
■ Double-click the **Regional and Language Settings** icon.	☐

ACTION	MRC
■ The popup window may be in Chinese if it was switched during the Windows XP installation process. Otherwise, the window will be in English. On the **Regional Options** tab, select “Chinese(PRC)” under **Standards and formats**; select “China” under **Location**.	☐

Fig. 66:



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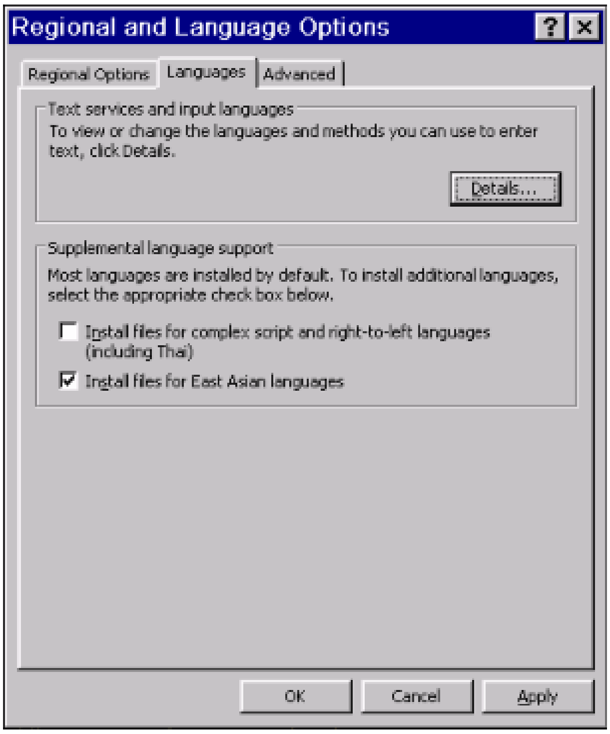
ACTION	MRC
■ Switch to the **Languages** tab:	☐

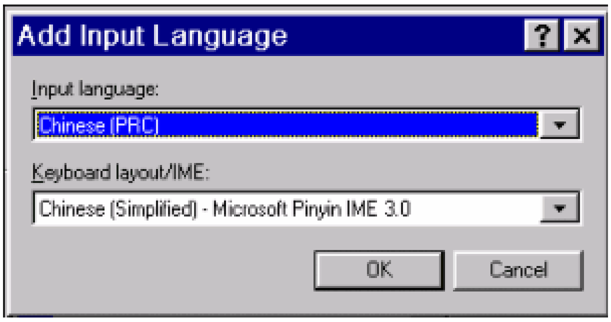
Fig. 67:



ACTION	MRC
▪ Click the **Details** button.	☐

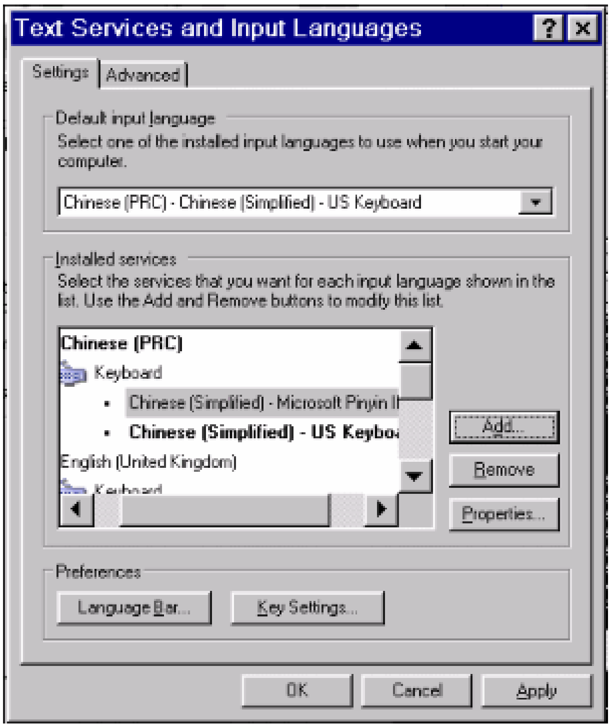
If the required keyboard layout/IME is not present, click the **Add** button, open the **Add Input Language** window, and select the appropriate input language and keyboard layout/IME.

Fig. 68:



Then click the **OK** button. The new settings shown below will apply:

Fig. 69:



ACTION	MRC
■ Click the **Language Bar** button under **Preferences**, check the two checkboxes, then click the **OK** button.	☐

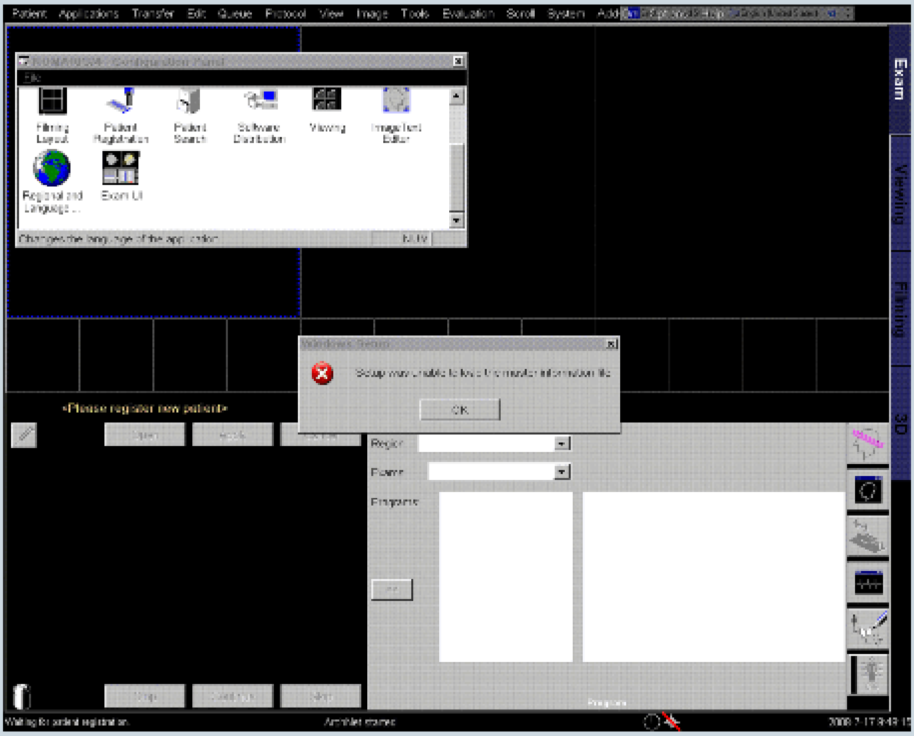
Fig. 70:

ACTION	MRC
■ Click the **OK** button in the **Text Services and Input Languages** window. Click the **Apply** button in the **Regional and Language Options** window. Do not click the **OK** button in the **Regional and Language Options** window at this point, as this will start an automatic reboot.	☐
■ Open Windows Explorer (it may be necessary to switch to advanced user settings to open the Explorer).	☐
■ Delete file C:\MedCom\config\Mri\MRSystem\MrVCheck.ref.	☐
■ Now click the **OK** button in the **Regional and Language Options** window, and confirm all messages with "YES" or "OK". A reboot will follow.	☐
■ Log in as meduser (if autologin is not enabled) after the reboot and wait until the main UI is displayed.	☐

When the language is switched to Chinese:

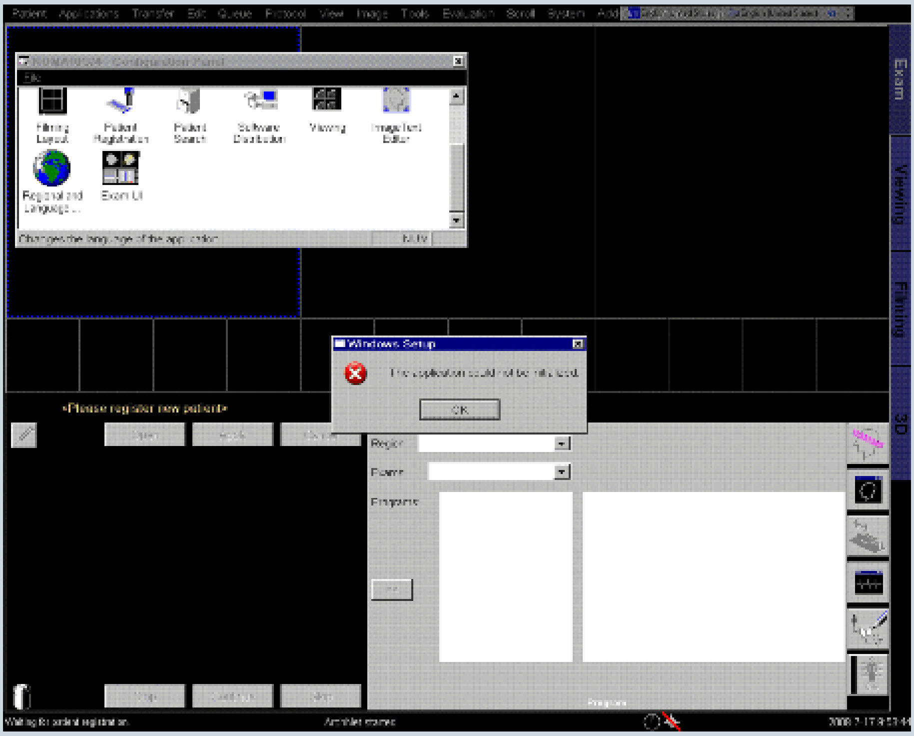
- If the error message “Setup was unable to load the master information file” appears, ignore it and click “OK”.

Fig. 71:



If the error message “The application could not be initialized” then appears, ignore it as well and click “OK”.

Fig. 72:



The system will then reboot. Once it has rebooted, the language will be switched to Chinese.

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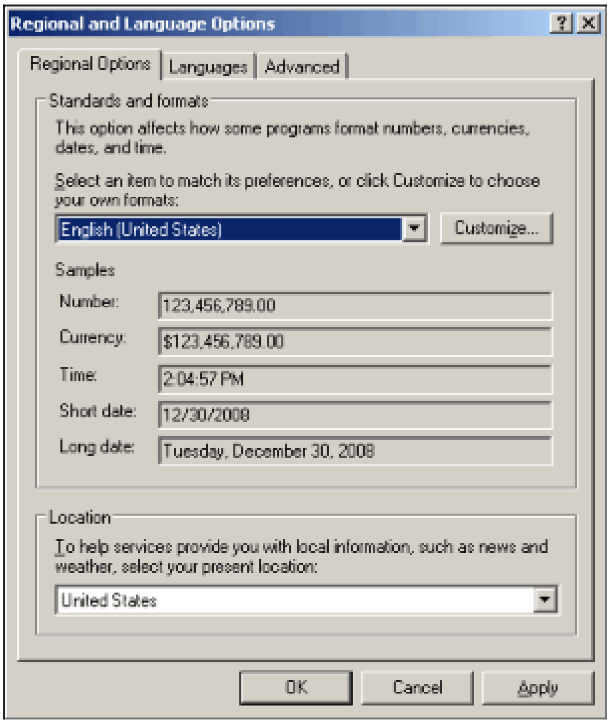
6.1.2.2 Switching the language from Chinese to English (US)

ACTION	MRC
▪ Log in as meduser. In the main UI, select **Options** -> **Configuration**.	☐
▪ Double-click the **Regional and Language Settings** icon.	☐
▪ On the **Regional Options** tab, change the settings to US English.	☐

Fig. 73:



ACTION	MRC
▪ Click the **Details** button. If the required keyboard layout/IME is not present, click the **Add** button, open the **Add Input Language** window, and select the appropriate input language and keyboard layout/IME. Then click the **OK** button. The process for making the new settings is now complete.	☐

ACTION	MRC
■ Click the **Language Bar** button under **Preferences**, check the two checkboxes, then click the **OK** button. *Fig. 74:* 	☐
■ Click the **OK** button in the **Text Services and Input Languages** window. Click the **Apply** button in the **Regional and Language Options** window. Do not click the **OK** button in the **Regional and Language Options** window at this point, as this will start an automatic reboot.	☐
■ Open Windows Explorer (it may be necessary to switch to advanced user settings to open the Explorer).	☐
■ Delete file C:\MedCom\config\Mri\MRSystem\MrVCheck.ref.	☐
■ Now click the **OK** button in the **Regional and Language Options** window, and confirm all messages with "YES" or "OK". A reboot will follow.	☐
■ Log in as meduser (if autologin is not enabled) after the reboot and wait until the main UI is displayed.	☐

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6.2 Archiving

6.2.1 Configuring the CD-R drive as an archive device

If the customer requires the CD-R drive to be an archive device, you can configure it as such in the Service Platform.

1. Start local service.
2. Enter the valid service key and click **OK** to confirm.
3. Select **Configuration**.
4. Under DICOM, select **Offline Devices**.
5. Select the CD-R entry under the **Select logical name** drop-down menu.
6. Check the **Archive device** box.
7. Click **Save** and **Finish**.
8. Reboot by clicking **Home**.
9. Confirm the windows displayed by clicking **OK**.

After the system reboots, the following entry will appear in the Viewer window: "Transfer, Archive to CD-R".

6.2.2 Settings for Codonix color printer NP1660M

6.2.2.1 Description:

Problems with the quality of the printout from Codonix Color printer NP1660M.

6.2.2.2 Workaround:

Check the settings of the printer if the configuration is set to color.

- Go to local service.
- Select Configuration -> DICOM Print devices.
- Go to step 2, HC device.
- Select the entry for the Codonix color printer under HC device.
- Check if "color appearance" is set to **Color 24 Bit** if printing is taking place on paper (medium type is set to paper).

  If medium type is set to blue film, the entry for "color appearance" must be set to **Grayscale 8 bit**.

  Change any incorrect settings.
- Don’t forget to save any changes made.

6.3 Troubleshooting

6.3.1 Customer network IP conflicts with measurement network

If the customer local area network uses the same IP address range as the measurement network (host, MPCU, and Imager), the addresses for the measurement network have to be changed. Refer to the following procedure:

- Start the local service from the mrc and select **Configuration**.
- Select **TCP/IP**.
- Select the network adapter "[00000001] Measurement (internal)". This NIC is used for the measurement network.
- Select the IP address and click **Del** to remove the original entry.
- Add the new IP address (e.g. 192.168.4.1) and the subnet mask (e.g. 255.255.255.248) by inserting them in the appropriate fields, and click **Add**.
- Click **Save** to save the changes.
- Reboot the mrc.
- Log in as "meduser".
- Shut down and switch off the system, wait ~20 seconds, and boot the system again.

The measurement network should now work with the new measurement network addresses.

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Version02 vs Version01

Chapters	Changed
Software installation procedure for the MRC	Update the correct software baseline

8 List of Hazard IDs

There are no Hazard IDs in this document.

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